



Eco-Sequencher

A Computer Engineering Practice and Design Project

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By:

Flores, Vincent R.

Franco, Ronniel R.

Legaspi, John Paul

Mendez, Eugene R.

Punzalan, James G.

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**APPROVAL SHEET**

This CPE Practice and Design Proposal entitled “**Eco-Sequencher**” proposed and submitted by Vincent R. Flores, Ronniel R. Franco, John Paul Legaspi, Eugene R. Mendez, and James G. Punzalan in partial fulfillment of the requirements for the degree **Bachelor of Science in Computer Engineering**, has been examined and found in order and is hereby recommended for acceptance and approval for **ORAL EXAMINATION**.

CPE PRACTICE AND DESIGN PROPOSAL COMMITTEE**Mr. Percian Joseph C. Borja MIT**

Chair

Ms. Jenice Anne Marie B. Visperas, MITc

Member

Engr. Jenny Rose D. Ventura

Member

PANEL OF EXAMINERS

Passed by the PANEL OF EXAMINERS on ORAL EXAMINATION on June 22, 2024.

Ms. Jenice Anne Marie B. Visperas, MITc

Chair

Mr. Percian Joseph C. Borja, MIT

Member

Engr. Saira Marie S. Mangayao

Member

Engr. Adelino B. Cordova Jr., MECpE

CpE Practice and Design Adviser

Accepted and approved in partial fulfillment of the requirements for the degree **Bachelor of Science in Computer Engineering**.

Date**Ms. Charmie Lynn N. Seno, LPT**

Proofreader

Date**Engr. Adelino B. Cordova Jr., MECpE**
Computer Engineering Department Head

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ABSTRACT

Plastic bottles, renowned for their unmatched convenience in storing beverages, have become indispensable in daily life. However, their extensive usage has led to a significant environmental crisis, particularly in the Philippines, where it ranks third globally in plastic waste generation, amounting to 2.7 million metric tons. Addressing this urgent challenge is essential for safeguarding the planet's health and securing a sustainable future.

To an extent, these significant issues also represent confirmed challenges for stakeholders at Eastwoods Professional College of Science and Technology Inc. In response to the acknowledged necessity, a defined set of technologies has been proposed, complete with details on schematics, codes, and implementation procedures.

This study demonstrates the Eco-Sequencher's System, this technology can mitigate the environmental concern posed by discarded plastic bottles, while also providing a source of clean and filtered water that benefits individuals seeking access to potable water without the need to purchase it.

Keywords: image processing; potable water; filtration; collection; plastic bottles; reverse vendo machine.

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CHAPTER I

THE PROBLEM AND ITS BACKGROUND

Introduction

This chapter discussed the Background of the Study, Statement of the Problem, Objectives of the Study, Scope and Delimitation, Significance of the Study, and Definition of Terms.

Background of the Study

The undeniable convenience that plastic bottles have brought into our daily lives, serving as essential containers for both hot and cold beverages. However, this widespread convenience has given rise to a compelling environmental concern – plastic pollution. Despite the undeniable utility they offer, the adverse impact on the environment, as emphasized by Parker (2019), has propelled us to address this global issue. This problem is particularly pronounced in the Philippines, ranking third in plastic waste generation worldwide, contributing 2.7 million metric tons, according to Jambeck et al. (2015).

Hence the consequences of plastic pollution extend globally, affecting marine ecosystems, and locally, especially in coastal areas like Bataan, where urban drainages contribute significantly to plastic waste in the ocean. The occurrence of this also results in a microplastic that can enter humans and animals when digested the fragments prevailed at two of the WTPs (Water Treatment Plan) and fibers together with fragments predominated in one case. Despite 12 different materials forming the microplastics being identified, the majority of the MPs (>70%) comprised of PET (polyethylene terephthalate), PP (polypropylene), and PE (polyethylene) which is present in plastic bottles, As stated by Janda et al. (2018) and their study contributes to fill the knowledge gap in the field of



emerging microplastic pollution of drinking water and water sources, which is of concern due to the potential exposure of micro-plastics to humans.

The dilemma faced by the individual who goes to Eastwoods Professional College of Science and Technology Inc. Every day in the vicinity of Barangay Ibayo Balanga City, Bataan. According to a conducted survey on January 26, 2024, 28 out of 30 students said that there is a lack of access to free potable water that the school offers since the free potable water from the dispenser is only available for the staff of EPCT and not to students, as well as plastic bottles that are not properly segregated in the trashcans. Ms. Doreliza Pearl R. De Guzman the administrator of the institution agreed that potable water is one of the basic needs of the students of EPCST.

The ease of use of reverse vending machines brings inspiration to researchers who are exploring innovative ways to enhance user experiences across various technologies and systems that interact with these machines in the process of returning and recycling items providing valuable insights into user behaviors, preferences, and the psychology behind sustainable practices. To address the major concerns of segregation of plastic bottles and the lack of potable water inside the institution, the researcher has finished designing a machine that allows individuals of Eastwoods Professional College of Science and Technology to exchange their thrown-away plastic bottles for a credit point that is inspired by a reverse vending machine to address the two issue with just one solution to quench their thirst with a clean and filtered source of potable water, so addressing both the issue of improper plastic disposal and the necessity for drinking clean water. This solution can minimize the dangers to the environment created by disposed plastic bottles, and the source

of clean and filtered water also benefits the individual who wants to drink clean water without spending money to buy one.

Statement of the Problem

This research project revolves around the deployment and assessment of Eco-Sequencher, which involves the sustainable transformation of plastic bottles into clean water sources to quench the thirst of everyone who uses this machine.

It aimed to address the following inquiries:

1. How can this device assist and handle the issue at Eastwoods Professional College of Science and Technology in terms of?
 - a) Collection of plastic bottles.
 - b) Providing a clean source of water.
2. What are the technologies to be used in creating the Eco-Sequencher in terms of?
 - a) Hardware.
 - b) Software.
3. What are the features and functions of Eco-Sequencher?
 - a) Proper Collection of Plastic Bottle.
 - b) Water Filtration System.
4. How would the Eco-Sequencher be evaluated and identify the benefits it offers about:
 - a) Perceived of Usefulness.
 - b) Perceived Ease of Use.

Objectives of the Study

This study's general objectives are the development and implementation of Eco-Sequencher.

Specifically, it aimed to:

1. Determine the user's ability to assess current practices.
 - a) The device seeks to optimize the proper collection of plastic bottles.
 - b) Ensure the provision of a clean water source by filtering water.
2. Identify and understand the various components used to create the Eco-Sequencher in terms of:
 - a) Hardware.
 - b) Software.
3. To optimize the collection efficiency, enhance filtration, and raise awareness through the implementation of the Eco-Sequencher at Eastwoods Professional College of Science and Technology, addressing plastic bottle disposal and providing clean water access.
 - a) Ensure that the collected plastic bottles are properly segregated and managed to reduce environmental impact.
 - b) Incorporate a robust and effective water filtration system that meets the standards for potable water.
4. Evaluate and identify the Eco-Sequencher benefits it can offer regarding.
 - a) Perceived of Usefulness.
 - b) Perceived Ease of Use.

Significance of the Study

As the result of this study, the project Eco-Sequencher aims to benefit the following:

Students of EPCST – can greatly benefit from Eco-Sequencher because it can provide a sustainable and convenient water source, encourages plastic bottle segregation, and instills a sense of environmental responsibility.

Local Communities – they can benefit from it to practice proper segregation of plastic bottles.

Environmental Advocates – they can benefit from it to do an initiative action to support broader public awareness and appreciation for eco-friendly initiatives, contributing to the global movement towards sustainability.

Future Researchers – offers valuable reference and inspiration for future projects, encouraging the ongoing development and improvement of eco-friendly technologies. By using it as a basis or as pertinent material for their research, the researchers may even use

Scope and Delimitation

The researchers created a research criterion focused on the development and implementation of Eco-Sequencher.

This part lays out the extent of the study, identifying the many outcomes that must be completed. It includes the entity.

1. Users insert a plastic bottle which is then compressed by a linear actuator upon acceptance.
2. Identifying the plastic bottle using a camera through image processing (Python and OpenCV).

3. The system detects plastic bottles with liquid content, which are not accepted. It uses an HX711 load sensor, with a minimum acceptance weight set 20 grams below the limit, as bottles undergo compression during processing, preventing water splatter within the machine.
4. A servo motor is used to accept and dispense valid plastic bottles for conversion. Users must manually retrieve bottles deemed invalid due to contents such as water, ensuring removal before continuing.
5. Four-channel relays control the linear actuator's movements (forward and backward), activate a solenoid valve for water, and operate a UV light sterilizer to conserve electricity.
6. Water passes through a four-stage filtration process, including 10-micron, 5-micron, and 1-micron filters, and UV light sterilization, ensuring it meets drinking water standards.

Definition of Terms

The following terminologies are defined operationally to get a better understanding of the terms used.

Cluster Sanitary Landfills - a method of waste disposal where multiple smaller landfills are used together to manage waste in a more organized manner.

Ecological Solid Waste Management Act - legislation focused on proper waste management to address environmental concerns related to solid waste.

Ecologically Sustainable Practices - strategies or actions that support environmental balance and minimize negative impacts on ecosystems.



Land-based Plastic Pollution - plastic waste originating from land sources, is often transported via waterways to marine environments.

PET (Polyethylene Terephthalate) bottles - a type of plastic commonly used for beverage packaging, known for its recyclability.

Reverse Vending Machine - a device that rewards users for returning used items, like plastic bottles, by providing incentives or credits.

Surface Run-off - water flows over surfaces due to precipitation, carrying plastic debris to rivers and subsequently to seas.

Waste segregation - the systematic sorting and separation of waste materials, such as plastic bottles, for proper recycling or disposal purposes.

CHAPTER II

REVIEW OF RELATED LITERATURE

In this chapter, the researcher delves into a comprehensive analysis of global plastic consumption, unraveling its multifaceted dimensions and implications. Focuses on the various literature evaluations and studies that inspired Eco-Sequencher. The researchers assemble all the information from multiple publications and journals on a website that includes a reference from the paper's author. This may help enhance the current study by analyzing the literature given. The following are the important components that need to be studied to support the Eco-Sequencher: The researchers compiled different topics from the gathered data that gave support and added value to the study, such as Linear Actuator, UV light, filtration, and reverse vending machine.

Linear Actuators

Similarly, as stated by Endaya et al. (2020), they focused on the design and implementation of an efficient waste segregation system. This system utilizes ultrasonic sensors to determine whether paper or plastic bins need compression and employs linear actuators to compress the waste, thereby maximizing the space available in the trash bins. The primary goal of this system is to prevent resource over-extraction and environmental degradation by improving waste management practices. The use of linear actuators in this context highlights their importance in enhancing system efficiency.

Further exploring the significance of linear actuators, Zhu et al. (2020) designed a direct-drive compressor using an electromagnetic linear actuator, aimed at enhancing compressor efficiency and meeting energy conservation and environmental protection requirements. Their study commenced with the structural design and performance analysis

of the linear compressor, followed by an examination of the working process of a moving-coil linear compressor. The basic performance of the designed moving-coil linear motor was simulated and analyzed using Maxwell software, while the overall performance of the linear compressor was verified through Simulink software simulations.

Additionally, Pereira et al. (2019) propose a novel approach for a leg-based stair-climbing mechanism. This approach focuses on a mechanical modification of a previously patented design, enhancing the mechanism's flexibility and enabling the wheelchair to move up and down without changing its orientation—an innovative feature not yet implemented in practice. Central to this advancement is the use of electric linear actuators in place of the originally proposed pneumatic system. Electric linear actuators provide improved control, efficiency, and ease of use, making them a superior choice for this application.

Moreover, Galik et al., (2020), plastics in discarded forms are pervasive in our surroundings, with their excessive production posing a significant current issue and an even more formidable challenge for the future. Despite being frequently utilized for their positive attributes, the drawbacks, such as extended decomposition time, are noteworthy. While the expectation is that separation, collection, and recycling are standard practices, they often prove elusive. Plastic waste management is notably inefficient due to the disproportionately large space occupied by plastics relative to their weight. Addressing this inefficiency necessitates the incorporation of crushers and waste volume reduction, particularly to mitigate logistical costs in plastic handling and preparation for subsequent recycling stages. This composition outlines a potential design for a plastic waste shredder,

detailing key components and providing a machine visualization. A critical aspect is the resolution of the crushing mechanism, a pivotal element in the equipment's operation, with the project undertaken at the Department of Design and Machine Elements.

Global Plastic Consumption

As reported by Walker & Fequet (2023), the escalating global plastic pollution issue and the subsequent formation of micro (Nano) plastics emphasize environmental threats and ecological impacts. Despite existing downstream strategies to address plastic pollution, the paragraph argues that they could be more effective due to the continued rise in plastic production and waste generation, surpassing current regulations. The international community acknowledges the need for a more comprehensive approach to tackle plastic and micro (Nano) pollution. The critical review underscores the lack of successful prevention or removal techniques for micro (Nano) plastic pollution while offering constructive criticism and insights into strategies that can contribute to achieving ambitious global reduction targets for a sustainable future in plastics.

Meanwhile, according to Wright et al., (2020), the degradation of plastic bottles, as highlighted in the study, leads to the formation of microplastics, which possess the potential to infiltrate water systems and subsequently enter the human food chain. This process raises serious concerns about the potential impacts on human health. The microplastics generated from the breakdown of plastic bottles can carry hazardous substances and chemicals, creating apprehension about the possibility of ingestion through contaminated water and food sources. The study underscores the need for heightened awareness and comprehensive measures to address the intricate relationship between microplastics, environmental contamination, and potential human health risks.

In addition, a study by Smith et al., (2019), plastic production has grown exponentially, with an estimated 9.2 billion metric tons produced between 1950 and 2017. However, 8–11 million metric tons of plastic waste escape into the oceans each year. The packaging industry accounts for 47% of plastic usage. Recycling and the circular economy are seen as key solutions, but the plastic problem continues to grow. More attention must be given to developing technologies for waste management, integrating international collaboration, and reducing waste at its source point. A legally binding global governance arrangement is needed to effectively limit and control plastic pollution. Many governments are in favor of this solution.

According to Gall & Thompson (2015), due to their affordability and ease, plastic products, such as bottles and bags, are widely used in both industrial production and daily life. They pose a threat to social progress and environmental preservation nonetheless, as their widespread use causes serious issues like resource waste and plastic waste contamination. According to Statistical Product and Service Solutions' sociological survey in Kinshasa, every tactic for lowering the pollution caused by plastic waste has a major positive influence. Government policy came in second at 42.3%, and environmental considerations came in first at 88.8%. At 36.0%, economic issues had the least effect. This research could provide a foundation for creating waste management plans or putting pollution control measures for plastic garbage into practice.

Furthermore, the study of Jambeck et al., (2015), the escalating issue of plastic bottle pollution is becoming increasingly worrisome. A significant proportion of discarded plastic bottles, as highlighted in their findings, find their way into aquatic environments, thereby intensifying the global plastic pollution crisis. This alarming trend poses a severe

threat to both marine ecosystems and terrestrial environments, as the persistence of plastic pollutants adversely impacts the delicate balance of these ecosystems. The study underscores the urgency of addressing and mitigating plastic bottle pollution to safeguard the health and sustainability of our planet's diverse ecosystems. The findings emphasize the critical need for proactive measures and responsible waste management practices to curtail the detrimental effects of plastic pollution on both aquatic and terrestrial habitats.

UV light

Based on the study by Ruiz-Diez et al. (2023), discusses the challenge of ensuring continuous access to drinking water in areas lacking municipal facilities for water sanitation. It highlights self-treatment as a realistic solution and introduces UV disinfection treatment as a portable alternative, particularly beneficial for remote, off-grid households. Its power requirements are the main obstacle to implementing UV disinfection in such settings. The paragraph introduces a portable water sterilizer device utilizing UV-C LEDs specifically designed for direct use in isolated rural communities. The study explores pulsed radiation operation as a strategy to reduce energy demand, and the Design of Experiments methodology is applied to optimize relevant parameters such as voltage, radiation time, frequency, and duty cycle. The goal is to enhance energy efficiency and effectively remove *Escherichia coli* from polluted water.

As stated by Nazar & Musawi (2021), ultraviolet (UV) rays are widely used for disinfecting microbial water, particularly when treating drinking water. UV disinfection is highlighted as a physical method that effectively kills bacteria and destroys cells, purifying water without generating toxic substances. The research aims to achieve environmentally sustainable drinking water in water scarcity areas. A pilot plant was established to assess

the efficacy of a ceramic filter in reducing turbidity and plankton, potentially altering drinking water properties. The research also aims to evaluate the effectiveness of UV sterilization, specifically in eliminating bacteria such as fecal coliform and *Escherichia coli*. This approach could streamline water purification stages and contribute to sustainable water treatment practices.

Along with that, S. Purwoto et al. (2020) discuss pollution in groundwater caused by pathogenic bacteria and parasites and suggest disinfection methods such as ozone, chlorine gas, and ultraviolet light to mitigate contamination. The study aims to assess the effectiveness of various treatments, including coagulation, filtration, absorption, deionization, and disinfection, in reducing parameters in groundwater to meet drinking water quality standards—the UV light treatment involved initial affixing a coagulant aid, followed by filtration using polypropylene sediments. Additionally, specific treatments were employed to remove iron (Fe) and manganese (Mn) using ferrite, decrease hardness using manganese greensand, lower cations using anion resins, and reduce anions using cation resins.

Moreover, Kai Song et al. (2016) found that UV disinfection is a highly efficient technology for deactivating waterborne pathogens and is gaining traction for industrial use. Over the last decade, a novel UV source, ultraviolet light-emitting diode (UV-LED), has emerged, offering several advantages over conventional UV mercury lamps. This promising alternative has sparked significant research on applying UV-LEDs for water treatment. In recent years, there has been a notable increase in studies exploring UV-LED water disinfection.

Filtration

As indicated by Cescon & Jiang (2020), the primary goal underscores the significance of filtration in the multi-barrier approach to drinking water treatment, particularly for pathogen removal. It explains that suspended solids and particulate matter can hinder disinfection, making efficient particle removal crucial for effective disinfection. Granular filtration, commonly sand-based, is highlighted for its potential to enhance disinfection efficiency. The paragraph also mentions the recent development and use of alternative filter media. The review focuses on comparing the performance of these alternative media with the traditional sand/anthracite combination in the context of drinking water treatment.

In line with this, Hube et al., (2020), highlight the promising potential of direct membrane filtration in wastewater treatment and resource recovery due to its superior treated water quality, efficient nutrient recovery, and sustainable operation. The review aims to provide a comprehensive overview of the current state of direct membrane filtration processes, including pressure-driven, osmotic-driven, thermal-driven, and electrical-driven methods. The focus is on treating various wastewater types for reclamation and resource recovery. The factors influencing membrane performance and treatment efficiency are discussed, with membrane fouling identified as a primary challenge. The paragraph emphasizes strategies to enhance direct membrane filtration performance, such as physical and chemical cleaning techniques and feed water pretreatment. Effective wastewater reclamation and resource recovery are emphasized in addressing the challenges and prospects for scaling up and achieving long-term operations.

Lastly, according to Evangelista et al., (2019), Filtration processes stand as pivotal solutions in contemporary water treatment endeavors. Given the prevailing deterioration of water quality owing to environmental degradation and industrial operations, ensuring water suitability for human consumption and domestic purposes remains imperative. In this context, a novel filtering cartridge is introduced within the discourse, seeking to address inherent shortcomings observed in conventional commercial counterparts. These deficiencies were elucidated through antecedent experimental endeavors conducted at the Laboratory of Environmental and Maritime Hydraulics (LIDAM) of the University of Salerno, with a specific focus on assessing head losses incurred by prevalent filtration cartridges, notably under conditions of progressive clogging. The adverse implications of head losses, particularly in the context of low-pressure systems, were discerned as potential impediments to the smooth operation of the installed apparatus. Through meticulous analysis of experimental findings, profound insights into the functioning of commercial filtration cartridges were gleaned, thereby culminating in the conception of a novel, economically viable, non-toxic, and enduring filtration cartridge. This innovative design paradigm features a central element comprising inert cotton encased within a cylindrical polypropylene framework, augmented by protective inclusions such as white marble pebbles, ceramic cylinders, and granular activated carbon.

Reverse Vending Machines

The study by Mariya et al., (2020), stated that effective waste management through recycling becomes crucial as the volume of generated waste rises amid limited landfill space. Plastic, a significant contributor to waste, is often discarded after use. The proposal involves the creation of a Reverse Vending Machine (RVM) with a reward system

designed explicitly for collecting plastic bottles. The technology employed for plastic bottle identification is image processing. Depositors can claim points by entering a unique ID, and these accrued points can be redeemed for promo codes for online shopping. Both users and authorities can monitor details by maintaining accounts in the developed application software.

Additionally, Gaur et al., (2018), recycling has emerged as a novel and practical waste management approach in response to the escalating waste generation and limited landfill capacity. The current recycling process, requiring individuals to transport bulk waste to recycling centers, can be inconvenient and discouraging. To address this issue, an automated recycle bin, derived from the reverse vending machine (RVM) concept, has been developed and deployed in various locations such as subways and malls. This innovative system incorporates a reward feature, utilizing fraud detection sensors to initiate recycling after inserting plastic materials. The plastic undergoes a thorough examination by a series of sensors. Users are incentivized with attractive rewards, receiving coins as a token of appreciation for utilizing the Reverse Vending Machine.

Based on Tomari et al., (2017), recycling is a crucial waste management strategy in response to the growing waste volume and limited landfill space. The current manual recycling practice, requiring users to transport bulk waste to recycling centers, can be inconvenient and discouraging. This project proposes an automated recycle bin with a reward feature inspired by the reverse vending machine (RVM) concept. The system is implemented in standard municipal recycle bins and incorporates a microcontroller and sensors for user identification and weight measurement. The sensors automatically convert the weight to corresponding points. Users can claim their points through an RFID point



card. A microcontroller controls the entire process. A small-scale user test demonstrates the system's effectiveness, aiming to motivate Malaysians to recycle and contribute to addressing urban poverty through a waste-to-wealth concept.

CHAPTER III

RESEARCH DESIGN AND METHODOLOGY

This chapter presents the research design and methodology used in this study including the methods, project development methodology, research respondents, and research instruments. The researchers use this method to discuss all the processes of Eco-Sequencher.

Development of the System

The functions and design of the Eco-Sequencher system are idealized from a series of conceptualizations. Furthermore, facts from the web, theories, books, interviews, and related studies have provided a basis for the developers and allowed them to acquire information and data. As the developers accomplished generating thoughts and identifying needs and requisites for the Eco-Sequencher System, they gained enough knowledge from previous developments and can address some of the questions regarding the issues. The developers utilized Agile Methodology, which is a flexible and iterative approach to project management, that emphasizes collaboration, users' feedback, and rapid, incremental delivery of software. In contrast to a linear-sequential life cycle model, Agile Methodology allows for overlapping phases and emphasizes user feedback and quick iterations. This approach accelerates the development process by focusing on rapid iteration cycles, enabling faster delivery of functional software and greater adaptability to changing requirements. This is supported by our use of Agile methods.

Research Methods

The researchers used both qualitative and quantitative methods in gathering information. The goal of the study is to collect and examine both numerical and non-numerical data to comprehend people's attitudes, beliefs, and motivations about their social reality. By using interviews, surveys, and observations, the researchers can comprehend and understand what issues need to be solved.

Utilizing qualitative research techniques, researchers may examine how people interpret both social and subjective phenomena in the larger picture of their everyday lives. Analyzing text involves a reflective process of systematic gathering, organizing, and comprehension, usually from conversations or observations according to Malterud, (2022). The researchers conducted an interview that was checked and validated by the adviser before proceeding and observation was used to evaluate and test the device

Research Locale

The researchers' proposal, which is the Eco-Sequencher study, was conducted at Eastwoods Professional College of Science and Technology Inc. at Barangay Ibayo Balanga, Bataan. As the locale for the proposed study, it was chosen since the institution lacks access to free potable water since the free potable water from the water dispenser is only for the staff of EPCST and not for the students; student bring their tumbler with water and some of them even buy from the convenience store or in retailer store nearby the institution. The proper practice of segregation of plastic bottles. These considerations are guided by the specific research objectives, the geographical reach of the Eco-Sequencher System, available resources, and the intended focus of the study. If the Eco-Sequencher project has been deployed in this area, the primary research site naturally centers on this

Eco-Sequencher

locale, allowing for the assessment of direct impacts on the community and the environment.

Respondents of the Study

The respondents of the study were the college students of Eastwoods Professional College of Science and Technology and Engineering Experts. Since the number of students is 942 the researcher uses the Slovin formula, employing a 10% margin of error to ensure a representative sample from the total population of 942 students. This calculation yielded a sample size of 90 students who participated in the research and added the 2 Engineering Experts providing insights into their perspective regarding the Eco-Sequencher. Refer to Appendix C for the total number of College Students.

This is the Slovin Formula:

$$n = \frac{N \cdot}{1 + N_e^2}$$

n = Sample size

N= Total population size

e = Margin of error expressed as a decimal fraction

$$n = \frac{942}{1 + 942 \times 0.10^2}$$

$$n = \frac{942}{1 + 942 \times 0.01}$$

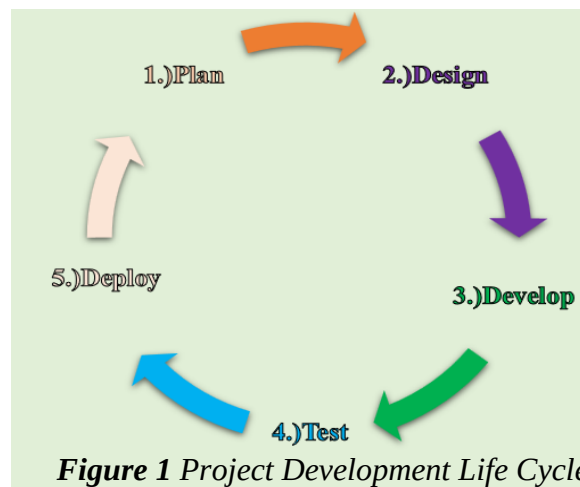
$$n = \frac{942}{1 + 9.42}$$

$$n = \frac{942}{10.42}$$

$$n = 90 + 2 = 92$$

Project Development Methodology

The proposed project employed the Agile Model, a popular choice among various SDLC Models because it facilitates iterative development and frequent testing to ensure that requirements are fully met before deployment. According to research by Williams and Cockburn (2023), Agile methodologies outperform traditional waterfall models in terms of project success rates. Data from the 2023 CHAOS Report by the Standish Group shows that 42% of Agile projects are completed, compared to only 14% of waterfall projects. Additionally, Agile projects have a failure rate of just 9%, whereas waterfall projects fail 29% of the time. These statistics are corroborated by numerous industry reports, underscoring the effectiveness and higher success rates of Agile methodologies in software development.



Five System Development Life Cycle of Agile Model

1. **Plan-** In this phase gather information on using the interview of personnel-in-charge that can answer the various tasks needed to perform to have a broad comprehension and define the features and functionalities needed for the project developed.

2. **Design**- The system design displays all needs, as well as the necessity based on the data acquired for the project's development. This covers the parts and software programs that are utilized for execution. The work done for the beneficiaries as it is being used is highly valued in the researchers' initiative.
3. **Develop** - This is where actual development for its software development is done. The software begins to take shape and develop into a finished tangible device.
4. **Test** - This is a continuous cycle in the development cycle to determine whether it meets the specified requirement, or the improvement is needed. Researchers conduct numerous tests throughout the development cycle to find and fix errors and problems. The aim is to guarantee reliable and accurate program performance.
5. **Deploy** - After completing the previous stages the developed product can serve its purpose and be available to the real-life environment.

Research Instruments

The researchers employed both qualitative and quantitative data collection methods to gain comprehensive insights into the multifaceted aspects of the Eco-Sequencher project. Quantitative methods included internet searches for statistical and informational data, while qualitative approaches involved conducting interviews with stakeholders to understand their perspectives. Additionally, direct observations of user interactions and environmental impacts were conducted to gather qualitative data. Systematic record-keeping was also utilized to document project-related data and information over time, providing both quantitative and qualitative insights into the project's progression and outcomes.

Internet Search - used this instrument for searches that provide a wide range of data, including statistics on plastic pollution and information from reports and case studies on similar technology initiatives.

Interviews - with stakeholders yield valuable insights into their perceptions, motivations, and experiences related to the project.

Observation - user interactions, environmental impacts, and machine operation provide insights into how the technology is used and its impact on the environment.

Tracking Record - of project-related documents, including technical specifications and user feedback, allows for systematic documentation and analysis of data and information over time.

Survey - conducted to gather quantitative data on public awareness and attitudes towards reverse vending machines and plastic recycling initiatives. The survey includes questions aimed at understanding the public's knowledge of environmental issues, their willingness to participate in recycling programs, and their perceptions of the effectiveness of reverse vending machines in addressing plastic pollution.

System Development Tools

Block Diagram - a graphical representation that depicts the functional interrelationships of components within a system, illustrating the flow of signals or information between these components. It uses blocks to represent system elements and lines or arrows to indicate the direction of signal or data flow, providing a high-level overview of the system's structure and operation.

Flow Diagram - A flow diagram is a graphical representation of a process, system, or workflow, showing the steps or activities involved and their sequence. It uses standardized symbols to depict different types of actions, decisions, inputs, and outputs, making it easier to understand and analyze the process. Flow diagrams are commonly used in various fields, including engineering, business, software development, and project management.

Wiring Diagram - A wiring diagram is a detailed schematic representation of an electrical circuit or system. It shows the components of the circuit, their connections, and the layout of the wiring. Wiring diagrams are essential tools for electricians, engineers, and technicians, as they provide a clear and organized way to understand the electrical system's design and function.

Picture Diagram - A picture diagram, also known as a pictorial diagram, is a visual representation of a system or process that uses images or illustrations of actual components rather than abstract symbols or lines. It aims to provide a more intuitive understanding of how the components of a system are connected and interact with each other. Picture diagrams are often used in educational materials, user manuals, and technical documentation to help people understand complex systems more easily.

3D Modeling - is the process of creating a three-dimensional representation of an object or scene using specialized computer software, enabling the generation of realistic virtual or physical models. As Geospatial (2019) points out, 3D mapping means the three-dimensional analysis of objects to map objects in the real world. It involves the manipulation of geometric shapes and structures to simulate the appearance and functionality of the intended object in a digital environment.

Technology to be Used

Hardware

1. **Servo Motor** - A servo motor is a rotary actuator or linear actuator that allows for precise control of angular or linear position, velocity, and acceleration. It consists of a suitable motor coupled to a sensor for position feedback. A servo motor typically includes a controller that manages the motor's behavior based on the feedback it receives, allowing for precise positioning and speed control.
2. **HX711 and Load Cell** - The HX711 and load cell are commonly used together in weighing applications to measure force or weight.
3. **Arduino Uno and Arduino Nano** - Arduino Uno and Arduino Nano are both microcontroller boards developed by Arduino. Cc, based on the AT mega microcontrollers. They are popular in electronics projects for their ease of use, versatility, and extensive community support
4. **Piezo Buzzer** - a small electronic device that produces sound when an electric current passes through it. It operates using the piezoelectric effect, where certain materials generate a voltage when mechanical stress is applied.

5. **Water Flow Sensor-** a device that measures the flow rate of water in a system, providing valuable data for monitoring and controlling water-related processes.
6. **Solenoid Valve-** an electromechanical device that controls the flow of fluids or gases by using a magnetic solenoid to open or close a valve.
7. **Relays** - an electrically operated switch that controls the flow of current in a circuit, often used to control high-power devices with a low-power signal.
8. **4x3 Keypad Membrane-** a flat, flexible keypad with 4 columns and 3 rows of buttons, commonly used for input in electronic projects and devices.
9. **Jump Wires-** short wires with connectors at each end, used to create electrical connections between components on a breadboard or in electronic circuits.
10. **AC to DC Converter Power Supply** - A device that converts alternating current (AC) to direct current (DC), providing a stable power source for electronic devices.
11. **Breadboard-** a solderless device used for prototyping electronic circuits, featuring a grid of interconnected metal clips that allow for the temporary placement and connection of electronic components without the need for soldering.
12. **PCB (Printed Circuit Board)-** a board made of non-conductive material with conductive pathways etched or printed on its surface, providing a rigid platform for mounting and interconnecting electronic components in a compact and organized manner. PCBs are fundamental in the production of electronic devices and systems.
13. **20x4 i2c LCD Monitor-** provides a graphical user interface for users.
14. **16x2 i2c LCD Monitor-** provides a graphical user interface for users.
15. **Pushbuttons:** Allows user interaction with the system.

16. **Linear Actuator** – A linear actuator is a device that creates motion in a straight line, as opposed to the circular motion of a conventional electric motor. Linear actuators are used in a wide variety of applications where linear motion is required, such as in machinery, industrial equipment, and consumer products.
17. **UV- Light Water Sterilizer** - to kill bacteria and other microorganism that can affect water quality.
18. **Sediment Filters** - These are designed to capture and remove sand, silt, dirt, and rust from water, making it purer and cleaner to drink.
19. **UDF Carbon Filter** - is a porous material that can adsorb impurities and contaminants. In a water filter, activated carbon is used to trap and remove organic compounds, chemicals, and some dissolved gases from water.
20. **CTO Carbon Filter** is designed to address issues related to chlorine, as well as improve the taste and odor of water.

Software

1. **Arduino IDE** - is a software platform used for programming and uploading code to Arduino microcontrollers. It provides a user-friendly interface for writing, compiling, and uploading code to Arduino boards.
2. **Python** - a high-level, general-purpose programming language. Its design philosophy emphasizes code readability with the use of significant indentation. This is usually used by machine learning researchers for the handling of plastic bottle identification.
3. **Integration Tools** - to link the hardware components with the software system.

4. **OpenCV** - is an open-source computer vision and machine learning software library. It provides a comprehensive set of tools for image processing, computer vision, and machine learning applications. OpenCV is written in C++ and also has bindings for various programming languages, including Python.
5. **AutoCAD** - is computer-aided design (CAD) software developed and sold by Autodesk. It is widely used for creating 2D and 3D drawings, models, and designs in various industries such as architecture, engineering, construction, manufacturing, and more.

Evaluation Tool

The technology acceptance model (TAM) was used by the researchers to assess the project's quality attributes and user acceptability. TAM is used by researchers because it is a data framework hypothesis that predicts how people come to recognize and employ an invention. This technology acceptance model is divided into two primary factors to determine the use of the technology to emphasize the user's perspective: perceived usefulness and perceived ease of use. The quality of the project must meet the established needs and demands of its various users.

Chapter IV

PRESENTATION, ANALYSIS, AND INTERPRETATION OF DATA

This chapter presents the data, including the specifications, framework, improvements, and programming of the system, used to maximize the potential of the Eco Sequencer System in addressing significant environmental concerns. Following this, an analysis of the findings is provided. This comprises implementing the steps of the chosen software development life cycle in the system. Furthermore, utilizing the Technology Acceptance Model, this phase of the study assesses the functionality, performance, accuracy, effectivity, security, and usability of the designed system for the students at Eastwoods Professional College of Science and Technology.

Traditional Process of How to Collect Plastic Bottles

The traditional process of collecting plastic bottles begins with individuals or organized groups gathering used bottles from households, streets, and businesses. After collection, bottles are sorted to separate recyclable from non-recyclable materials and then cleaned to remove contaminants. Clean bottles are compacted or baled to facilitate transportation, which typically involves trucks or other means to deliver them to recycling facilities. At these facilities, bottles are shredded, melted, and molded into new products such as containers or fibers. The process ensures that plastic waste is diverted from landfills, promoting environmental sustainability through recycling and reuse.

Development of Eco Sequencher System

The functions and design of the Eco-Sequencher system are idealized from a series of conceptualizations. Furthermore, facts from the web, theories, books, interviews, and related studies have provided a basis for the developers and allowed them to acquire information and data. As the developers accomplished generating thoughts and identifying needs and requisites for the Eco-Sequencher System, they gained enough knowledge from previous developments and can address some of the questions regarding the issues. The developers utilized Agile Methodology, which is a flexible and iterative approach to project management, that emphasizes collaboration, users' feedback, and rapid, incremental delivery of software. This approach accelerates the development process by focusing on rapid iteration cycles, enabling faster delivery of functional software and greater adaptability to changing requirements. This methodology comprises several phases, beginning with the planning phase, followed sequentially by the design, development, testing, and deployment phases.

Planning Phase

Figure 2 shows that On January 9, 2024, the researchers conducted a comprehensive interview at Eastwoods Professional College of Science and Technology. The interview was held with Ms. Doreliza Pearl R. De Guzman, the Administrator of the School, who provided valuable insights that were meticulously recorded and gathered by the researchers. During the interview, Ms. De Guzman shared her perspectives and experiences regarding the ongoing projects and the potential impact of the proposed Eco-Sequencher System. She emphasized the importance of such initiatives in addressing environmental concerns and enhancing sustainability practices within the institution.



Figure 2 Actual Interview with School Administrator

Figure 3 1.1 shows the interview transcript that comprises detailed discussions with Ms. Doreliza Pearl De Guzman, providing valuable insights into their perspectives, experiences, and suggestions for enhancing the effectiveness of the project named Eco-Sequencher.

Our project is a machine with the capability to recycle or segregate plastic bottles in exchange for potable filtered clean water. For example, if you insert plastic bottles into our machine, it will identify if it is a plastic bottle and allow you to use the inserted plastic bottle in exchange for clean water using our machine. Additionally, you can store points using your student ID card to accumulate funds. If you need a source of clean water, you don't have to buy from retail stores anymore. This will quench your thirst and also help reduce the pollution caused by the improper disposal of plastic bottles, which is harmful to our environment.

1. Do you always consume potable water that is stored in a plastic bottle? If yes, do you buy it?

-Not really.

2. Do you think water should be free since it can be seen anywhere? If yes or no, why?

-It could be, on our campus, we need to have free water as a basic necessity.

3. Are you aware of the issue of plastic bottle pollution in our country? If yes, give your opinion about it.

-Yes, we really need to reduce the use of plastic bottles.

4. What do you think is a strategy we can use to raise awareness about the effects of plastic bottles and promote their exchange for clean water using the machine we propose?

-Absolutely.

Figure 3.1 Interview Transcript

5. In your opinion, will our proposed project help students get clean drinking water while also helping to segregate plastic bottles?

-It depends, but I think it's possible.

6. As an administrator of this school, would you want to implement this project if we get funding for it?

-If it turns out well, why not?

7. Do you also think it would help our maintenance to properly dispose of plastic bottles?

Yes, of course, because it will still be the students.

8. If our project is implemented in our school, will you use it instead of buying from outside? Why?

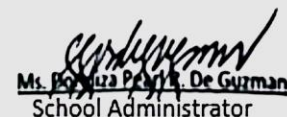
-That would be difficult, but we'll see if the school can provide the need.

9. As one of the managers of our school, do you think we can implement this project in our local barangay for it to become a project of our barangay officials?

-Because water is a need, I maintain that I think it cannot be done.

10. Give your opinion or suggestion about the problem that you want to be solved in our school.

My only suggestion is really about maintenance.



Ms. Bonanza P. De Guzman
School Administrator

Figure 3.2 Interview Transcript

On January 23, 2024, the researchers conducted online research as part of the planning phase for the Eco-Sequencher System. This research was essential for acquiring a comprehensive understanding of the requirements and functionalities needed for the project. The researchers utilized Agile Methodology, which emphasizes collaboration, user feedback, and rapid incremental delivery of software. This approach allowed them to

gather valuable information from various sources, including web theories, books, interviews, and related studies. The data collected provided a solid foundation for the development of the Eco-Sequencher System, enabling the researchers to address key environmental concerns and enhance the system's effectiveness and usability as shown in Figure 4 below.

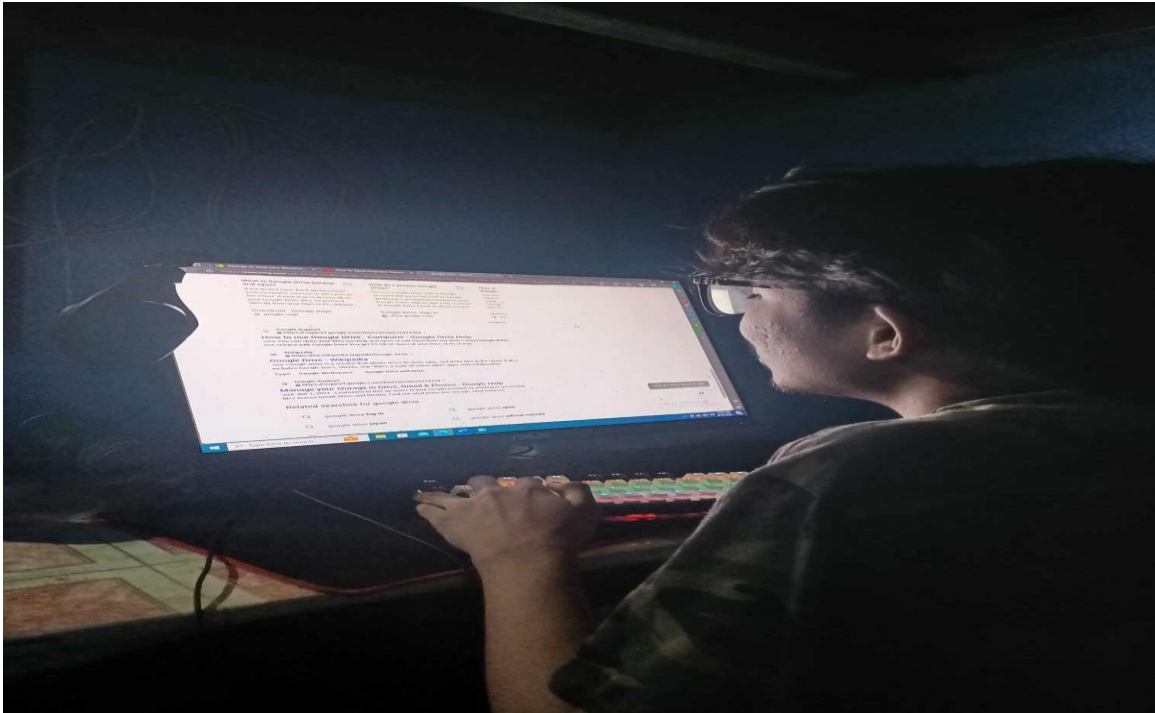


Figure 4 Online Research

Table 1 1.1 shows This project timeline outlines a series of milestones and tasks for the development of a project, divided by months from January to May. In January, the focus is on the proposal, analysis, and design, including identifying materials and technologies. February involves canvassing and selecting components, planning assemblies, and beginning construction. March is dedicated to testing and prototype creation, with tasks such as testing the water flow sensor and coding components. April continues with more testing, refinement, and adjustments, including training the camera

and conducting user tests. Finally, may focus on wiring management, design adjustments, and final fittings, leading to the project's completion.

Table 1 1.1 January to May Milestone

MILESTONE	TASK	DATE
1.0	Proposal, Agreement, and Interview	January
1.1	Analysis and Design Stage	
1.2	Identifying Possible Materials and Technologies to be Used	
1.3	Canvassing for the materials of the Chassis	
1.4	Plan for the Connection of the Project	
2.0	Canvass and Selection of Components and Hardware Materials	February
2.1	Plan for the Positions of Materials and Technologies	
2.2	Plan the Assembly of the Chassis	
2.3	Evaluation of Materials that fit for the Chassis	
2.4	Purchase the Materials for the Chassis	
2.4	Execute Plan Assembly of the Chassis	
2.6	Paint the Chassis and Secure the PVC Fittings	
3.0	Test Running the Pipes and Carbon Filters	March
3.1	Testing Water Flow Sensor	
3.2	Creation of Prototype	
3.3	Testing Prototype	
3.4	Conducting subsequent interview	
3.5	Coding for the components	
3.6	Testing of components	
2.9	Assembly of tube to the Chassis	
3.0	Implementation of Components on the Chassis	
4.0	Training the Camera	April
4.1	Purchase necessary things	
4.2	Adjustments	
4.3	Updating Papers	
4.4	Refinement	
4.5	User testing	
4.6	Adjustment	
4.7	Calibration	
5.0	Coding	
5.1	Wiring Management	May
5.2	Photoshop front of the chassis	
5.3	Printing the photoshop design	
5.4	Fitting the design	

Table 1 1.2 shows the final phase of the project, which is scheduled for June, and involves thorough testing and evaluation. This begins with the deployment of the project (6.0), followed by testing the entire project with the target beneficiaries (6.1). Based on the testing results, adjustments were made as needed (6.2). The project then undergoes an evaluation to assess its features and functionality (6.3). Continuous observation was conducted to monitor performance (6.4), and finally, the project officially concluded (6.5). This ensures the project meets its objectives and serves its intended purpose effectively.

Table 2 1.2 May to June Milestone

6.0	Deployment of the Project	June
6.1	Testing the Whole Project to the Target Beneficiaries	
6.2	Adjustments to the Project	
6.3	Evaluation of the Projects Features and Functionality	
6.4	Observation	
6.5	End of Project	

Design Phase

In this phase, after planning, the next step is to conceptualize the system's structure and functionality. This includes defining the overall architecture of the project and specifying hardware materials, software requirements, and functionalities.

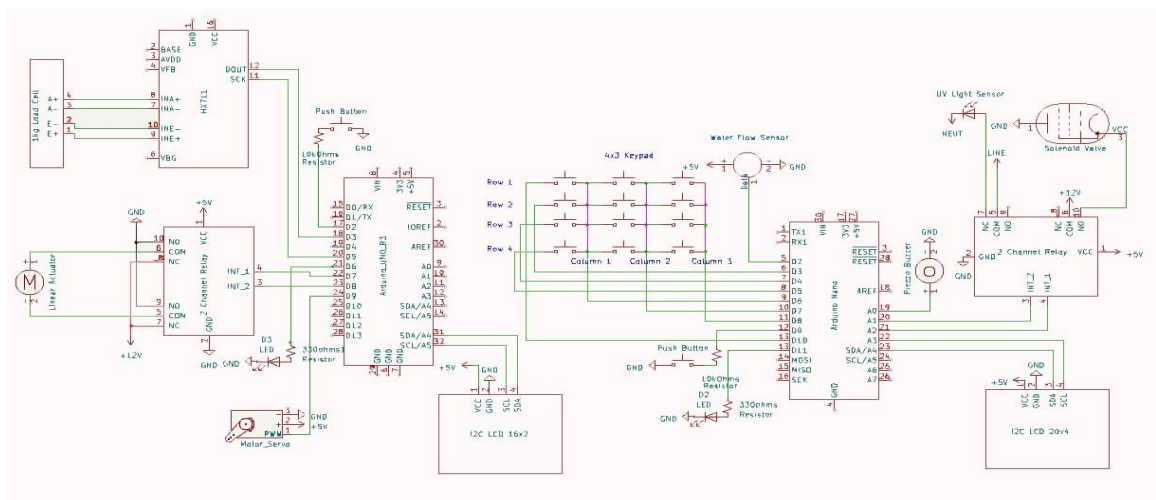


Figure 5 Wiring Diagram of Eco-Sequencer

Figure 5 The schematic shows the electronic circuit for the Eco-Sequencher system, featuring a microcontroller that coordinates components like a motor actuator for bottle compression, push buttons and a 4x3 keypad for user input, a water flow sensor to monitor water dispensing and a UV light sensor for ensuring water sterilization. Other elements include an LCD for the user interface, relays for controlling high-power devices, and a solenoid valve to regulate water flow.

Figure 6 shows the flowchart outlines the operation of the "Eco-Sequencher," a machine designed to recycle plastic bottles and dispense filtered water. Users begin by inserting a plastic bottle, which is verified by a camera and weighed to ensure it's empty. If the bottle meets the criteria, an actuator compresses it. Following this, users can select the amount of water they want via a keypad. An ultrasonic sensor detects if a container is placed for filling. If confirmed, the machine activates relays to turn on a UV light and open a solenoid valve, dispensing the selected amount of potable water. Finally, the machine thanks the user, and the process ends.

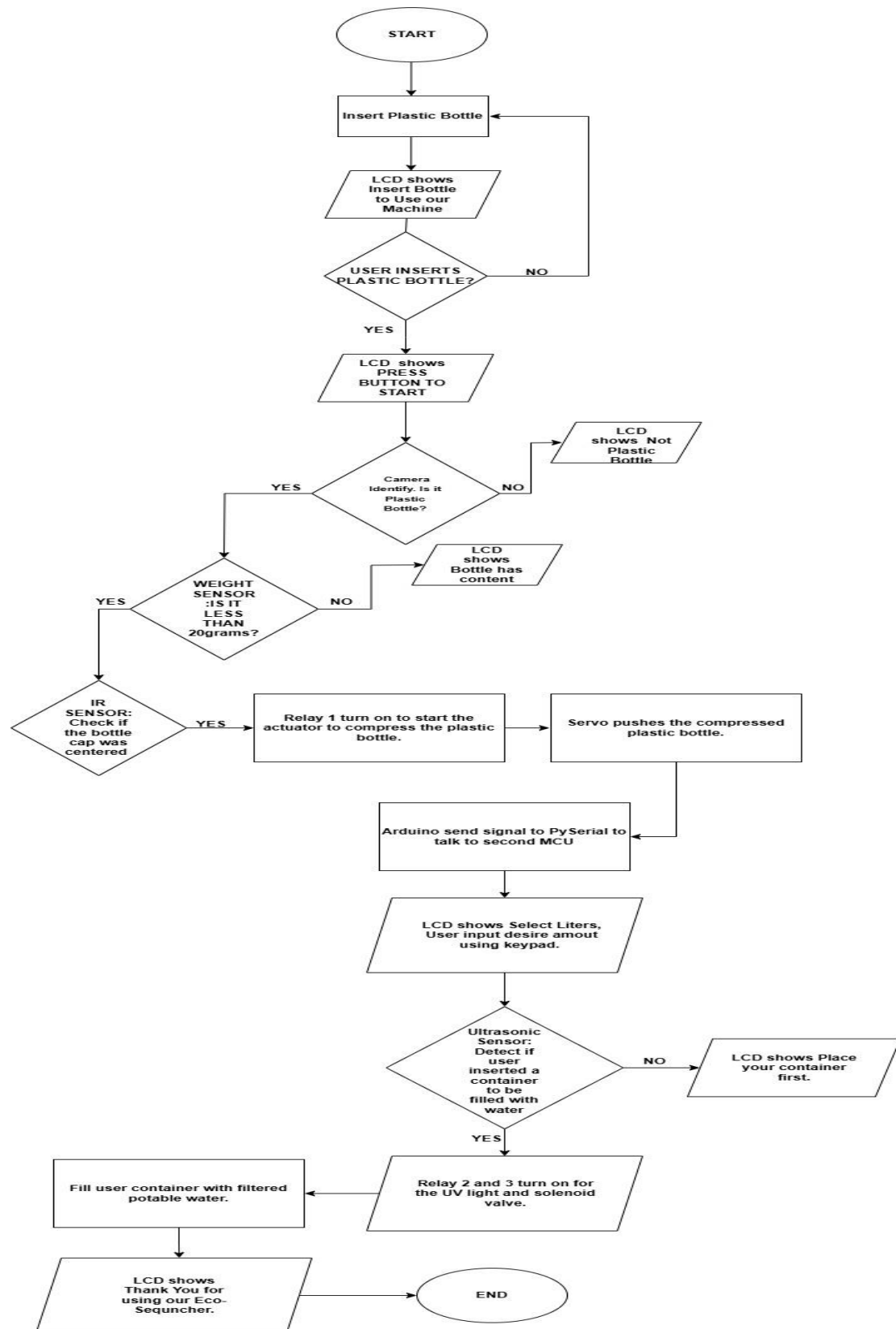


Figure 6 Flow Chart of Eco-Sequencer

Figure 7 shows the Picture Diagram which is a complex electronic system centered around an Arduino microcontroller. The Arduino coordinates various components, including a 4x4 keypad for user input, multiple relay modules for controlling high-power devices, and an LCD for output. Additional components include a 12-volt linear actuator, a water flow sensor, a power supply, and various switches and sensors. The relay interface between the Arduino and high-power devices, ensures safe and effective operation, while the keypad allows user interaction with the system. The LCD provides real-time feedback, making the system interactive and user-friendly.

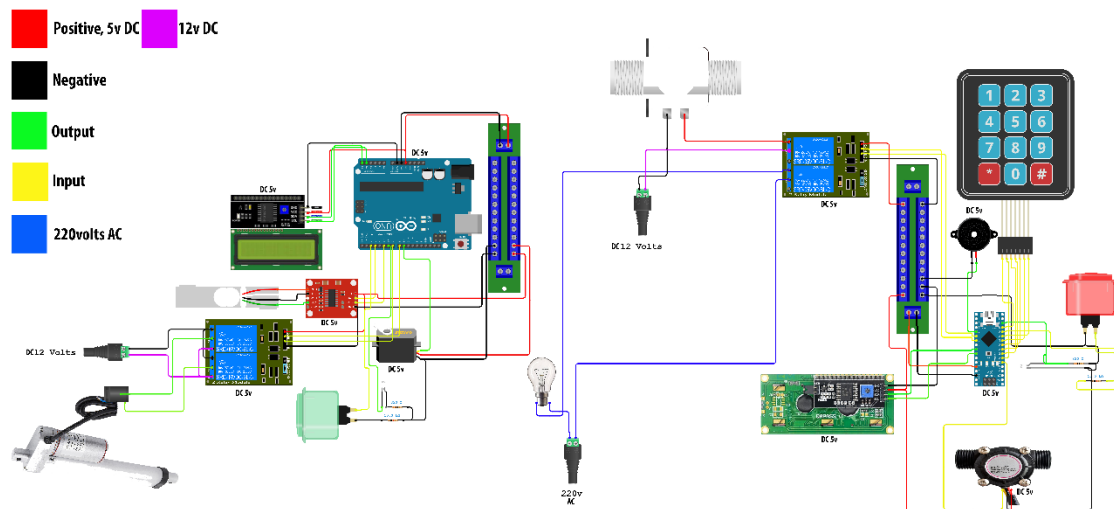


Figure 7 Picture Diagram of Eco-Sequencher

In Figure 8 1.1 It shows here the back view of the Eco-Sequencer that was created with the use of TinkerCAD and it shows 2 doors the left door is for the plastic bottle container which also is where we can access all the components inside for maintenance purposes.

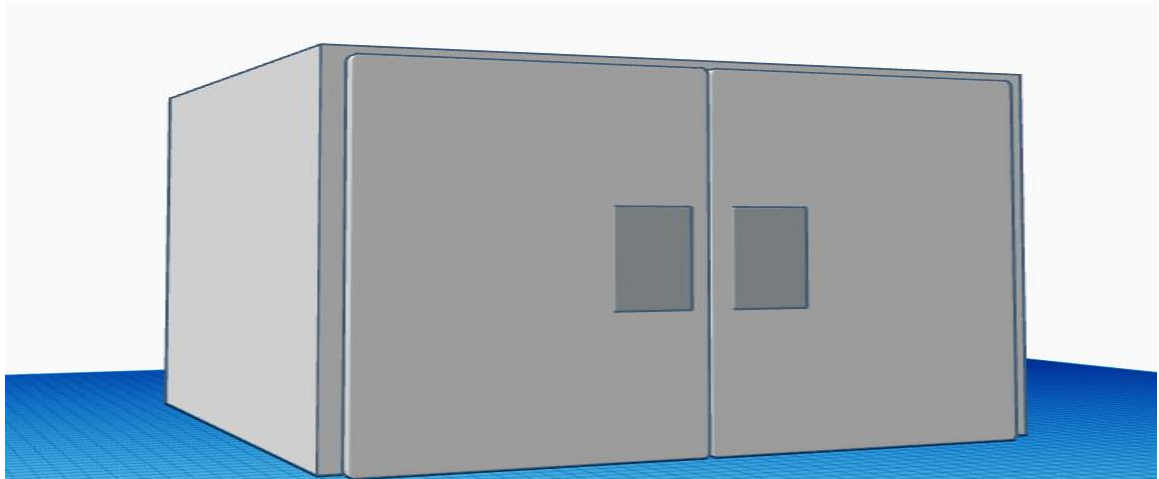


Figure 8 1.1 Back View of The Eco-Sequencer

Figure 8 1.2 shows here the front view of the machine and it shows the actuator that compresses the plastic bottle and sinks where you place your bottle or glasses to fill with clean and filtered water and the 4x3 Keypad Membrane which you can type how many ml do you want that seen on the LCD and also the red push button that indicates the stop and the green one is the start.

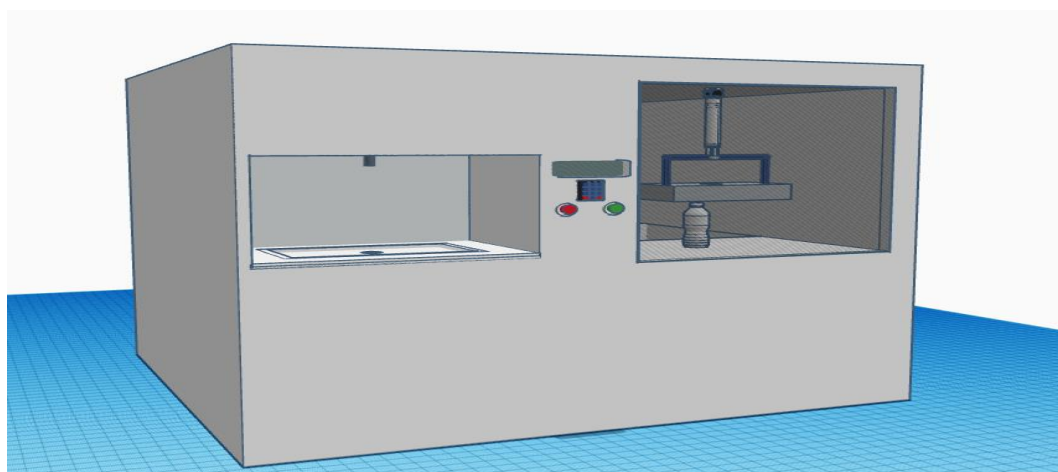


Figure 8 1.2 Front View of The Eco-Sequencer

Development Phase

The requirement specifications from the first phase are studied in this phase and the system design is prepared. This system design helps in specifying hardware and system requirements and helps in defining the overall system architecture.

In Figure 9 it shows here the Front view of the actual project, which is the Eco-Sequencher. You can see here that everything is ready and in the front view, there are instructions on how to use the machine and how it helps the environment.



Figure 9 Actual Front View of Eco-Sequencher

In Figure 10 the side view of the Eco-Sequencer shows that it aims to address plastic bottle waste by using image processing and machine learning to identify and compress bottles, which are then exchanged for clean drinking water through a filtration system and UV light. This initiative seeks to promote sustainability and environmentally friendly behaviors within the community by educating on plastic bottle recycling and fostering a culture of innovation and environmental responsibility.



Figure 10 Side View of The Actual Project

Parts and Functions of the Eco-Sequencher System:

Figure 11 shows the servo motor that helps move in precise, accurate, and controlled positions and rotations. For flicking the compressed plastic bottles inside the machine after it was accepted by the sensors and the camera.



Figure 11 Servo Motor

Figure 12 shows the Water Flow Sensor that helps measure the flow rate of water in the system, providing valuable data for monitoring. It helps the Arduino to read the pulse with modulation of it sends a signal to relay to stops which has the solenoid valve to close the valve so the water flow will stop.



Figure 12 Water Flow Sensor

Figure 13 shows the Solenoid Valve which is a device that controls the flow of fluids by using a magnetic solenoid to open or close a valve to stop the water from flowing once the desired input from the user is reached with the help of the water flow sensor.

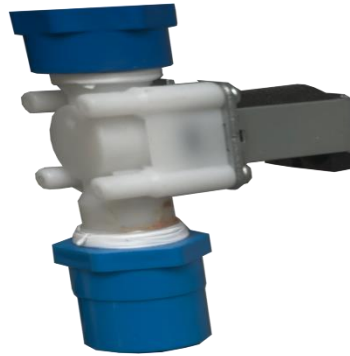


Figure 13 Solenoid Valve

Figure 14 shows the IR (Infrared sensor) that detects if the plastic bottle is placed in the center of the hole where the linear actuator mechanism can be seen.



Figure 14 IR Sensor

Figure 15 shows the Relay which is an electrically operated switch that controls the flow of current in a circuit, and is used to control devices with a low-power signal. Use to control the forward and backward motion of the linear actuator using 2 channel relays. The other two were used for the on and off the solenoid valve as well as the UV light sterilizer.

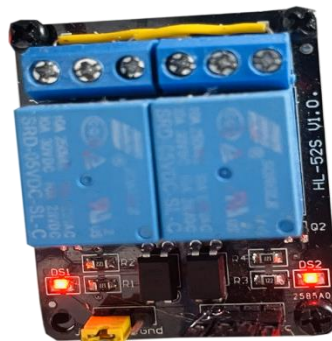


Figure 15 Relay Module

Figure 16 shows a 4x3 Keypad Membrane that is a flat, flexible keypad with 4 columns and 3 rows of buttons. Use for getting input from users depending on the amount of water they need.



Figure 16 4x3 Keypad Membrane

Figure 17 shows the Jump Wires which is a short wire with connectors at each end, used to create electrical connections between components on a breadboard or in electronic circuits.

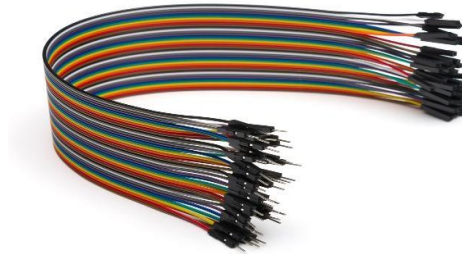


Figure 17 Jump Wires

Figure 18 shows the AC to DC Converter Power Supply which is a device that converts alternating current (AC) to direct current (DC), providing a stable power source for electronic devices. Use as the source for getting 12 volts for the linear actuator and the solenoid valve.



Figure 18 AC to DC Converter Power Supply

Figure 19 shows the Buck Converter DC to DC which is a type of DC-DC converter that steps down voltage, it was used to lower down voltage from 12v to 5v to ensure enough power is regulated.

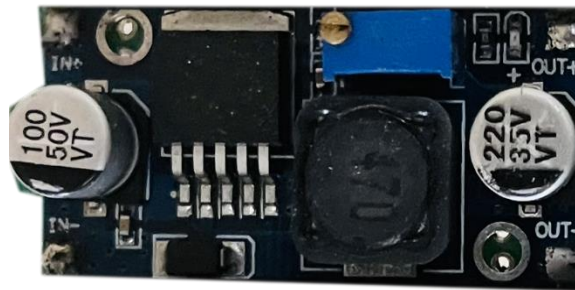


Figure 19 Buck Converter DC to DC

Figure 20 shows the Breadboard that is used for prototyping each component and to test if it is working. These holes are connected internally in a specific way, usually according to a standard layout.

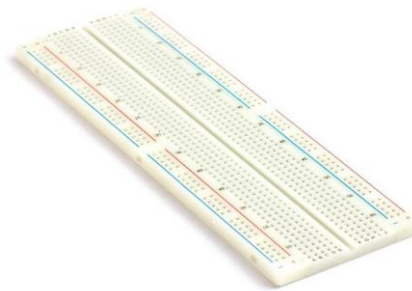


Figure 20 Breadboard

Figure 21 The PCB it is used to distribute power from Arduino to get 5 volts and grounds.

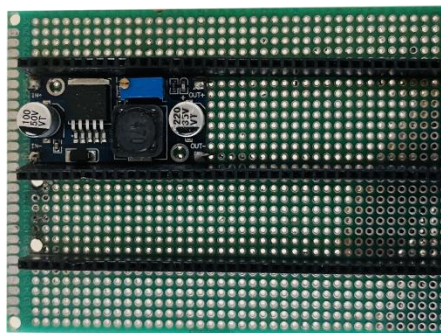


Figure 21 PCB (Printed Circuit Board)

Figure 22 shows the 20x4 i2c LCD Monitor that provides a graphical user interface for users. To interact with the user about what is happening with the system.



Figure 22 20x4 i2c LCD Monitor

Figure 23 The Push Button Allows user interaction with the system. To start and to stop.



Figure 23 Push Button

Figure 24 shows the Actuator that creates motion in a straight line, and it is used by Eco-Sequencher to compress the size of plastic bottles to save space.



Figure 24 Linear Actuator

Figure 25 shows the UV-Light Water Sterilizer that is used to kill bacteria and other microorganism that can affect water quality.



Figure 25 UV- Light Water Sterilizer

Figure 26 The Sediments Filter are designed to capture and remove sand, silt, dirt, and rust from water, making it purer and cleaner to drink.



Figure 26 Sediments Filter

Figure 27 shows the UDF Carbon Filter which is a porous material that can adsorb impurities and contaminants. In a water filter, activated carbon is used to trap and remove organic compounds, chemicals, and some dissolved gases from water.



Figure 27 UDF Carbon Filter

Figure 28 shows the CTO Carbon Filter that is designed to address issues related to chlorine, as well as improve the taste and odor of water.



Figure 28 CTO Carbon Filter

Figure 29 shows The Arduino R3 that was used as the microcontroller board in handling the plastic compression with the help of image processing to shrink the size of the plastic bottle controlling the relay for the linear actuator as well as the servo for the acceptance of plastic bottle flicking inside the machine.

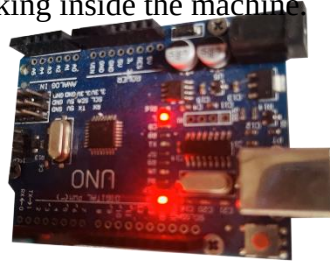


Figure 29 Arduino Uno R3

Figure 30 shows The Arduino Nano which is a small, versatile microcontroller. Was used for handling the refill process for the water, with the help of serial commutation we communicated the PC and the two-microcontroller unit.

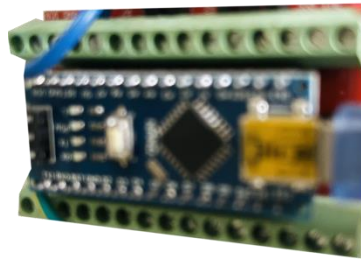


Figure 30 *Arduino Nano*

Figure 31 shows the 1080 p camera that is used for identifying plastic bottles that were inserted into the system using Python and OpenCV for image processing.



Figure 31 *1080p Camera*

Figure 32 shows the ultrasonic sensor which is a device that uses ultrasonic sound waves to measure distances. It works on the principle of sending out a sound wave at a frequency above the range of human hearing, typically around 40 kHz, and then listening for the echo when the sound wave reflects off an object.



Figure 32 *Ultrasonic Sensor*

Figure 33 shows A filtering housing that is often referred to as a filter housing or filter housing unit, is a container or casing that holds a filter cartridge or filter element. Its primary function is to protect the filter cartridge and facilitate the filtration process.



Figure 33 Filtering Housing

Testing Phase

In this phase, the System of the researchers is tested to identify defects, ensure it meets specified requirements, and validate its functionality, performance, and security. The main objectives are to ensure that the water is dispensed properly and accurately to confirm and fulfill user needs. The goal is to ensure the software is reliable, functional, and ready for deployment, contributing to overall quality and user satisfaction.

The researcher thoroughly checked all aspects of the technology to ensure that there were no flaws in the system or difficulties with the wiring also to ensure an accurate amount of water that the Eco-Sequencher released, the researchers conducted 7 days of testing shown in the table below.

Table 2 shows the accuracy of the amount of water input and output over a 7-day testing period. On day 1 the researchers input 250 ml and the water output was 3Liter which is not accurate, until day 4 it was still not accurate but when day 5-7 came it became accurate after many tries.

Table 3 7 days Calibrating Water Accuracy

Day	Time	Amount of water input	Amount of water outputs
1	11:00 am	250 mL	643 mL
2	8:00 am	500 mL	894 mL
3	5:00 pm	250 mL	673 mL
4	3:00 pm	250 mL	534 mL
5	11:00 am	250 mL	349 mL
6	12:00 pm	250 mL	298 mL
7	7:00 am	250 mL	250 mL

Researchers use decision tables to provide an organized method of listing all potential circumstances and related actions. This ensures that all possible scenarios are considered and methodically examined.

Decision Table

Legends

R: The plastic bottle is recognized.

NR: The plastic bottle is not recognized.

C: The bottle compression process is initiated.

NC: The bottle compression process is not initiated.

Table 3 shows the Basic Plastic Bottle Processing decision table

Table 4 Basic Plastic Bottle Processing

Condition	Rule 1	Rule 2
Recognition of the bottle	R	NR
Bottle compression	C	NC

Legends

W20: 20 grams below, accepted

W20A: 20 grams above, not accepted

Table 4 shows the Weight and Deformation Criteria decision table.

Table 5 Weight and Deformation Criteria

Condition	Rule 1	Rule 2
Weight	W20	W20A
Deformation	D	ND

Legends

PB: Plastic bottles, accepted

SD: Slightly deformed, accepted

FD: Fully deformed, not accepted

NPB: Not plastic bottles, not accepted

Table 5 shows the Plastic Bottle Conditions decision table.

Table 6 Plastic Bottle Conditions

Condition	Rule 1	Rule 2
Plastic bottles	PB	NPB
Deformation	SD	FD

Interpretation

Basic Plastic Bottle Processing

Case 1: Proper Bottle Processing

- Scenario: A user inserts a plastic bottle into the Eco-Sequencher.
 - Recognition: The system recognizes the bottle (R).
 - Compression: The system initiates the bottle compression process (C).
 - Interpretation: The system successfully processes the bottle and compresses it, ensuring efficient waste management and recycling.

Case 2: Improper Bottle Processing

- Scenario: A user inserts an item that is not recognized as a plastic bottle into the Eco-Sequencher.

- Recognition: The system does not recognize the item as a plastic bottle (NR).
- Compression: The system does not initiate the compression process (NC).
- Interpretation: The system fails to process the item and does not compress it, maintaining the integrity of the recycling process by ensuring only valid plastic bottles are processed.

Weight and Deformation

Case 1: Accepted Weight

- Scenario: A plastic bottle weighing 20 grams or below is inserted into the Eco-Sequencer.
 - Weight: The bottle is within the acceptable weight range (W20).
 - Interpretation: The system accepts and processes the bottle, ensuring efficient waste management and recycling.

Case 2: Not Accepted Weight

- Scenario: A plastic bottle weighing more than 20 grams is inserted into the Eco-Sequencer.
 - Weight: The bottle exceeds the acceptable weight range (W20A).
 - Interpretation: The system does not accept the bottle, ensuring only valid weights are processed.

Plastic Bottle Conditions

Case 1: Proper Plastic Bottle

- Scenario: A proper plastic bottle is inserted into the Eco-Sequencer.
 - Condition: The bottle is a valid plastic bottle (PB).

-Interpretation: The system accepts and processes the bottle, ensuring efficient waste management and recycling.

Case 2: Slightly Deformed Plastic Bottle

- Scenario: A slightly deformed plastic bottle is inserted into the Eco-Sequencher.

-Condition: The bottle is slightly deformed but still recognizable as a plastic bottle (SD).

-Interpretation: The system accepts and processes the bottle, ensuring efficient waste management and recycling.

Case 3: Fully Deformed Plastic Bottle

- Scenario: A fully deformed plastic bottle is inserted into the Eco-Sequencher.

-Condition: The bottle is fully deformed and not recognizable as a proper plastic bottle (FD).

-Interpretation: The system does not accept the bottle, ensuring only valid plastic bottles are processed.

Case 4: Non-Plastic Item

- Scenario: An item that is not a plastic bottle is inserted into the Eco-Sequencher.

-Condition: The item is not a plastic bottle (NPB).

-Interpretation: The system does not accept the item, maintaining the integrity of the recycling process by ensuring only plastic bottles are processed.

Deployment Phase

The deployment phase involves delivering the completed machine to the user and ensuring it operates correctly in its environment. This includes planning the deployment, installing and configuring the software for the machine, conducting initial and validation

tests, providing the user with a manual on how to use the machine, and offering post-deployment support. This phase ensures user readiness, system stability, and customer satisfaction while mitigating risks and facilitating a smooth transition to the deployment site.

Figure 34 shows the deployment letter signed by Ms. Doreliza Pearl R. De Guzman. This serves as proof of the school administrator's approval to deploy the project at their location. In this phase, the project was set up in the given area, and the monitoring system was tested to further check the functionalities and features of the project.

Dear Madam/Sir,

We hope this message finds you well. We, the undersigned researchers involved in the "Eco Sequencer" project, are writing to formally request authorization to proceed with the deployment phase of our innovative initiative.

The proposed deployment timeline covers one month (30 days), during this time we aim to implement the following features and functions of Eco Sequencer:

- Conversion of plastic bottles into clean water, offering a sustainable solution to plastic waste.
- Utilizing image processing, we have trained the camera through Python and OpenCV, enabling it to identify plastic bottles efficiently.
- Utilization of servo motors to accept and dispense plastic bottles that are valid for conversion, ensuring smooth operation and user convenience.
- Implementation of a four-stage filtration system, including 10-micron, 5micron, and 1micron filters, along with UV light sterilization, to ensure the purity of the converted water.
- Personalized filling of users' containers according to their preferences, based on their accumulated credit points, enhancing user satisfaction and engagement.

Additionally, we will require a dedicated space within the school premises to set up the Eco Sequencer. Furthermore, we will hire an expert in setting up water pipes (plumber) to ensure proper integration of our system with the existing infrastructure.

We also request access to the school's electricity and water supply to power the Eco Sequencer during its operation.

We are dedicated to ensuring the successful execution of the deployment phase and are prepared to promptly address any challenges that may arise.

Thank you for considering our request.

Noted by:


Engr. Adelino B. Cordova Jr.

CPE Practice and Design Adviser

Confirm by:


Ms. Dolores Pearl Orgezman

School Administrator

Sincerely,

Flores Vincent

Franco Rommel

Legaspi John Paul

Mendez Eugene

Punzalan James



Figure 34 The deployment letter

The researcher deployed the Project to Eastwoods Professional College of Science and Technology to be tested by the students. Before the beneficiary used the Eco- Eco-Sequencher System, the researchers discussed and demonstrated how to use the technology, so that, the Students at Eastwoods Professional College could evaluate it fully.

Table 6 shows How many users, how much amount of water is used, and how much the compressed plastic bottles of the Eco-Sequencher are in a span of 7 days. Additionally, In Appendix B you can see the actual photo of the users of Eco-Sequencher in a span of 7days

Table 7 Device Usage Information

	Users	Amount of Water Used	The weight of The Plastic Bottle Compressed	Number of Plastic Bottles inserted
Day1	9	2,250 mL	200 grams	10
Day2	8	2,000 mL	180 grams	9
Day3	8	2,000 mL	160 grams	8
Day4	6	1,500 mL	120 grams	6
Day5	9	2,250 mL	180 grams	9
Day6	5	1,250 mL	100 grams	5
Day7	7	1,750 mL	160 grams	8
Total:	52	13,000 mL	1,100 grams	55

Table 7 shows the respondents of the study. The Students of Eastwoods Professional College of Science and Technology. Additionally, 2 Engineering Experts were chosen for their ability to provide valuable insights and validate the findings, contributing to a comprehensive analysis of the technology's acceptance and effectiveness.

Table 8 Profile of the Respondents

Type	Total
Students	90
Engineering Experts	2
Total	92

The researchers used the Technology Acceptance Model (TAM) in the project to systematically understand and predict user acceptance and adoption of the new technology being introduced. TAM provided a clear and well-validated framework that allowed for the assessment of critical factors influencing user behavior, specifically perceived usefulness and perceived ease of use. By employing TAM, the researchers could identify and address potential barriers to adoption, ensuring that the technology was both useful and user-friendly from the perspective of the target audience. The model's extensive empirical support and adaptability to various contexts made it an ideal choice for the project, enabling the researchers to tailor their approach based on reliable predictors of user acceptance. This strategic use of TAM helped in designing better interventions, enhancing user satisfaction, and ultimately achieving higher adoption rates.

This section presents the results of the system evaluation. Ninety-two (92) respondents evaluate the system using Technology Acceptance wherein the respondent rate

Eco-Sequencher

the project using two primary factors: Perceived Usefulness and Perceived Ease of Use on a scale of 1-6 which are: strongly disagree, disagree, slightly disagree, slightly agree, agree, and strongly agree. The selected respondents to perform the evaluation are the students of Eastwoods Professional College of Science and Technology and Engineering experts. Table 8 shows the 6-point Likert scale that the researchers utilized in the study.

Table 9 6 Point Likert Scale

Weighted Mean Range	Rating Based on Quality	Weight
1.00 – 1.82	Strongly Disagree	1
1.83 – 2.65	Disagree	2
2.66 – 3.48	Slightly Disagree	3
3.49 – 4.31	Slightly Agree	4
4.32 – 5.14	Agree	5
5.15 – 6.00	Strongly Agree	6

Statistical Analysis of Data

On a 6-point scale, the weighted average of the 97 answers is 4.69. meaning that most of them agree with the TAM's two fundamental criteria for evaluating the Eco-Sequencher. The researchers conducted a comprehensive statistical analysis of data collected from Engineers, Students, and Environmental experts to investigate attitudes, preferences, and behaviors related to Eco-Sequencher adoption. Figure 30 shows the professions of the respondents who evaluated Eco-Sequencher.

Table 9 shows that in this study, the researchers integrated the use of bar infographics to effectively illustrate the data and to further enhance the visual

comprehension of the key findings of the ratings of 92 respondents on the Eco Sequencher System's perceived usefulness using a 6-point scale Technology Acceptance Model.

Table 10 *The general mean of Perceived Usefulness*

Criteria	Rating N=92												
	6	%	5	%	4	%	3	%	2	%	1	%	GWM
PERCEIVED OF USEFULNESS													
The Eco-Sequencher provides valuable benefits to the environment.	34	37%	30	33%	11	12%	16	17%	1	1%	0	0	4.54
Using the Eco-Sequencher is a cost-effective way to obtain clean water.	14	15%	35	38%	27	29%	15	16%	1	1%	0	0	4.5
Having access to clean water through the Eco Sequencher promotes preventive healthcare.	29	32%	37	40%	20	22%	4	4%	2	2%	0	0	4.94
The Eco-Sequencher is an eco-friendly device or machine.	23	25%	19	21%	43	47%	7	8%	0	0	0	0	4.63
The Eco-Sequencher is a reliable solution for segregating waste.	28	30%	29	32%	29	32%	4	4%	2	2%	0	0	4.83
The presence of Eco-Sequencher encourages the students to engage in discussions about water quality and environmental issues.	17	18%	38	41%	29	32%	6	7%	2	2%	0	0	4.67
The Eco-Sequencher helps the students save money by giving them access to clean water just by using an empty bottle.	29	32%	30	33%	14	15%	17	18%	2	2%	0	0	4.72
The Eco-Sequencher shows how small actions can help the environment.	25	27%	21	23%	26	28%	17	18%	3	3%	0	0	4.52
GENERAL MEAN BASED ON PERCEIVED USEFULNESS													4.67
Where: 6 – Strongly Agree, 5 – Agree 4, – Slightly Agree, 3 – Slightly Disagree, 2 – Disagree, 1 – Strongly Disagree													

Figure 35 shows that According to the bar infographic for the first criterion of perceived usefulness in Figure 0, 37% of respondents strongly agree, 33% agree, and 12% slightly agree that the Eco-Sequencher provides valuable environmental benefits. However, 17% of respondents slightly disagree, and 1% disagree. These results indicate that most respondents are convinced of the Eco-Sequencher's positive impact on the environment.

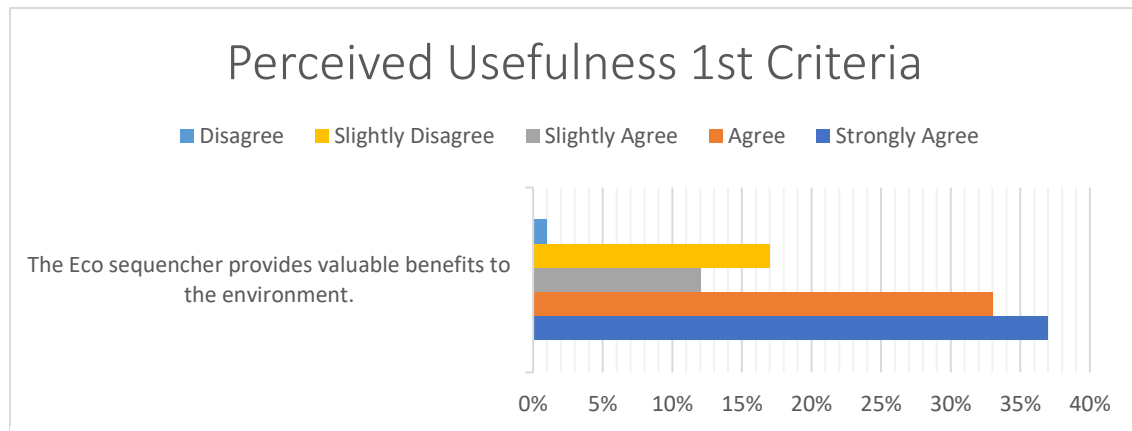


Figure 35 Perceived Usefulness 1st Criteria

It shows on Figure 36 shows the bar infographic for the second criterion of perceived usefulness, demonstrating that 15% of respondents strongly agree, 38% agree, and 29% slightly agree that the Eco-Sequencher is a cost-effective way to obtain clean water. Conversely, 16% of respondents slightly disagree, and 1% disagree. These results indicate that most respondents affirm the cost-effectiveness of using the Eco-Sequencher to obtain clean water.

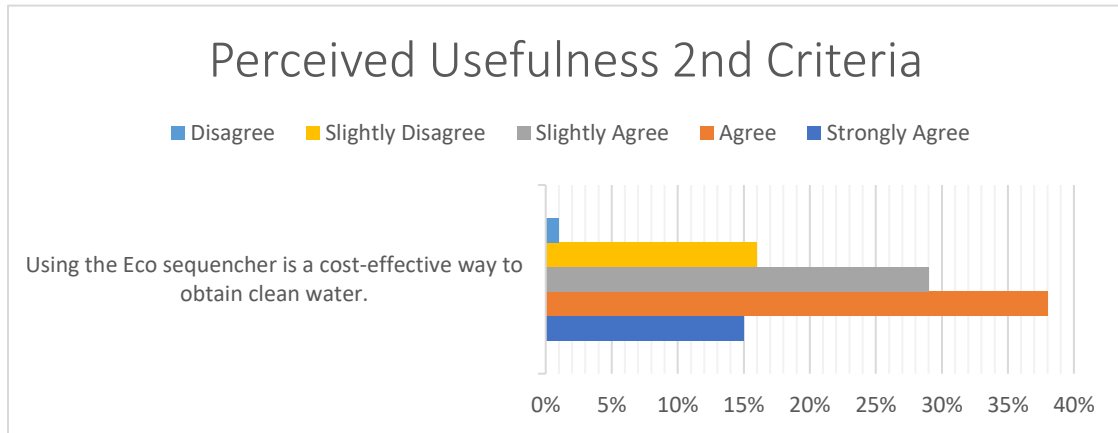


Figure 36 Perceived Usefulness 2nd Criteria

It shows on Figure 37 presents the bar infographic for the third criterion of perceived usefulness, indicating that 32% of respondents strongly agree, 40% agree, and 22% slightly agree that having access to clean water through the Eco-Sequencher promotes preventive healthcare. Conversely, 4% of respondents slightly disagree, and 2% disagree. These findings demonstrate that most respondents confirm the Eco-Sequencher's role in promoting preventive healthcare.

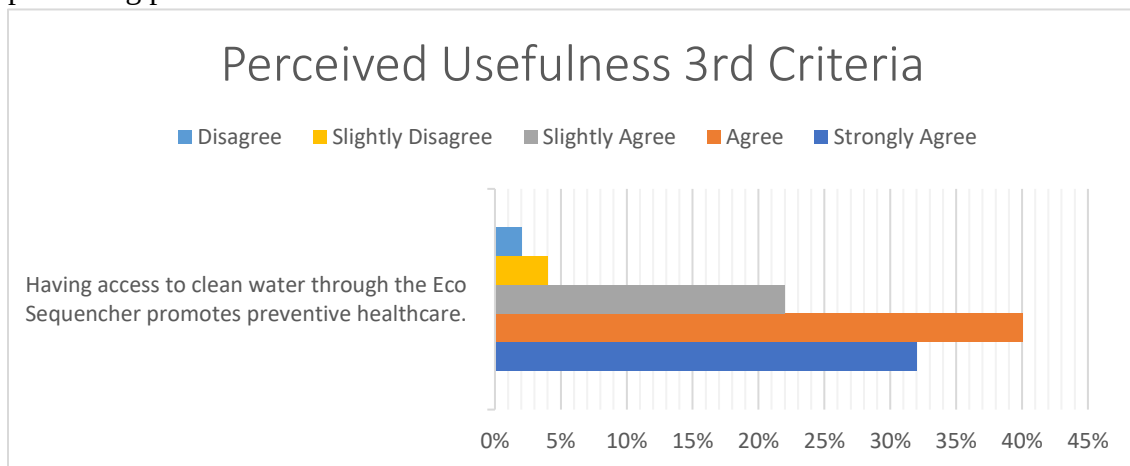


Figure 37 Perceived Usefulness 3rd Criteria

It shows on Figure 38 presents the bar infographic for the fourth criterion of perceived usefulness, indicating that 25% of respondents strongly agree, 21% agree, and 47% slightly agree that the Eco-Sequencher is an eco-friendly device. Conversely, 8% of

respondents slightly disagree. These findings demonstrate that most respondents are convinced of the Eco-Sequencher's eco-friendly nature.

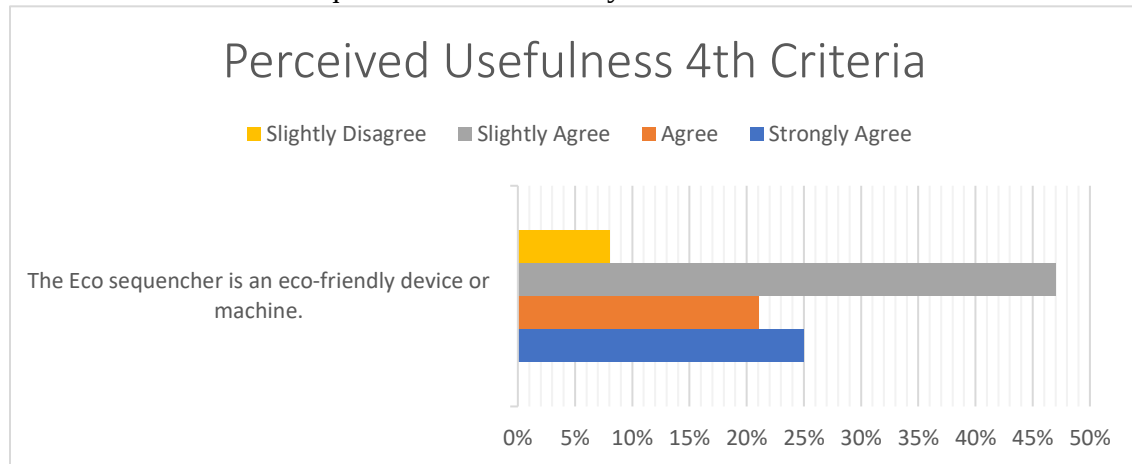


Figure 38 Perceived Usefulness 4th Criteria

It shows on Figure 39 shows the bar infographic for the fifth criterion of perceived usefulness, demonstrating that 30% of respondents strongly agree, 32% agree, and 32% slightly agree. Conversely, 4% of respondents slightly disagree, and 2% disagree. These findings indicate that most respondents certify that the Eco-Sequencher is a reliable solution for waste segregation.

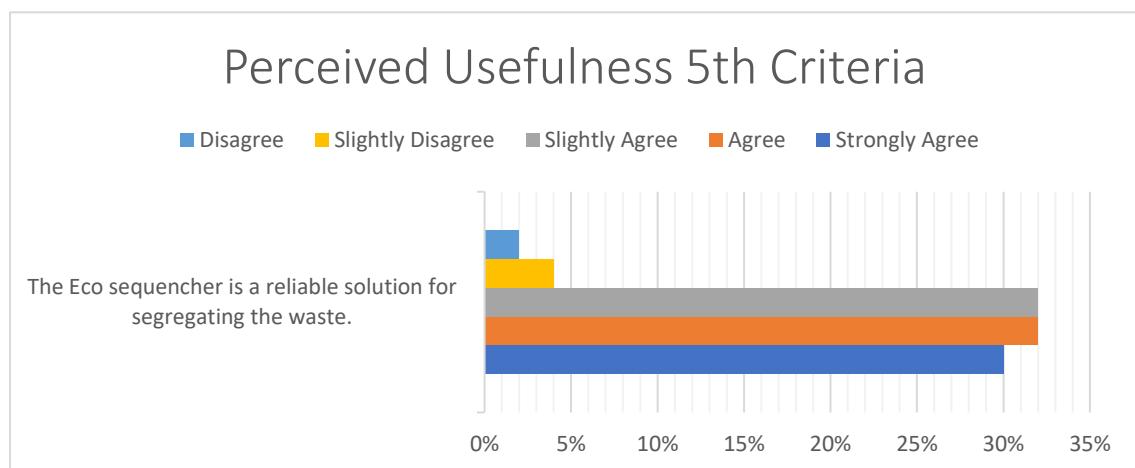


Figure 39 Perceived Usefulness 5th Criteria

It shows on Figure 40 shows the bar infographic for the sixth criterion of perceived usefulness, demonstrating that 18% of respondents strongly agree, 41% agree, and 32% slightly agree. Conversely, 7% of respondents slightly disagree, and 2% disagree. These findings indicate that most respondents confirm that the presence of the Eco-Sequencher encourages students to engage in discussions about water quality and environmental issues.

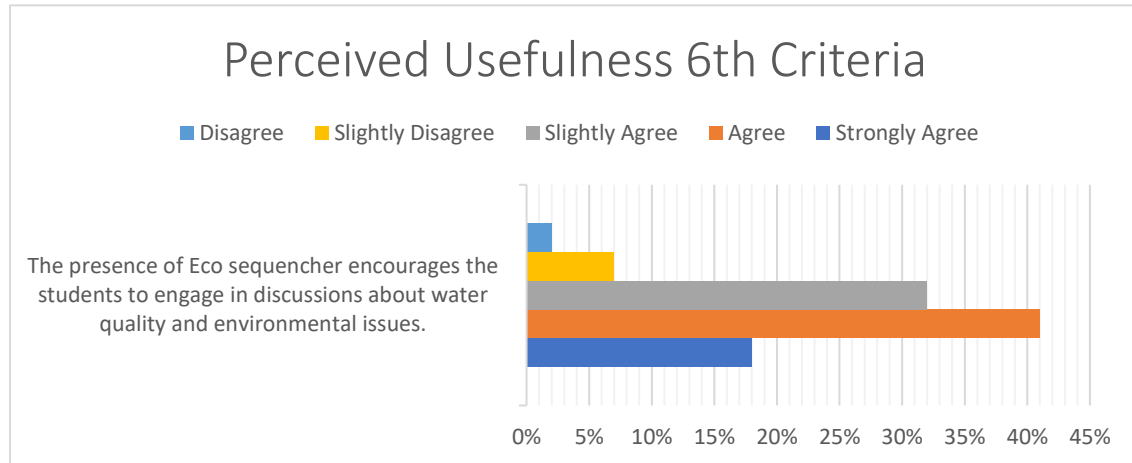


Figure 40 Perceived Ease of Use ^{8th} Criteria

It shows on Figure 41 presents the bar infographic for the seventh criterion of perceived usefulness, demonstrating that 32% of respondents strongly agree, 33% agree, and 15% slightly agree that the Eco-Sequencher helps students save money by providing access to clean water using an empty bottle. Conversely, 18% of respondents slightly disagree, and 2% disagree. These results indicate that most respondents affirm that the Eco-Sequencher helps students save money by providing access to clean water with just an empty bottle.

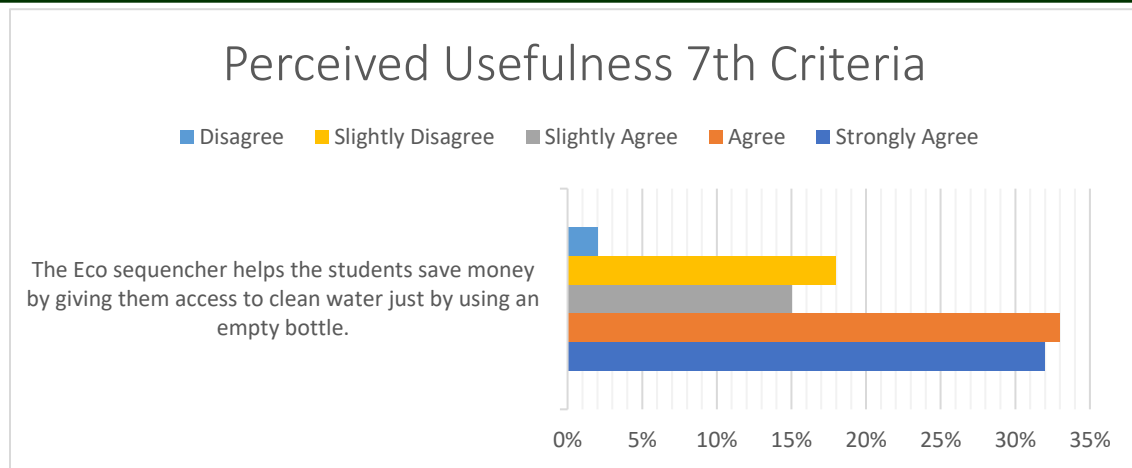


Figure 41 Perceived Usefulness 7th Criteria

It shows on Figure 42 presents a bar infographic illustrating responses to the eighth criterion of perceived usefulness regarding the Eco-Sequencher. The results indicate that 27% of respondents strongly agree, 23% agree, and 28% slightly agree that the Eco-Sequencher demonstrates how small actions can benefit the environment. Conversely, 18% of respondents slightly disagree, and 3% disagree with this statement. These findings suggest that most respondents affirm the Eco-Sequencher's effectiveness in showing how small actions can positively impact the environment.

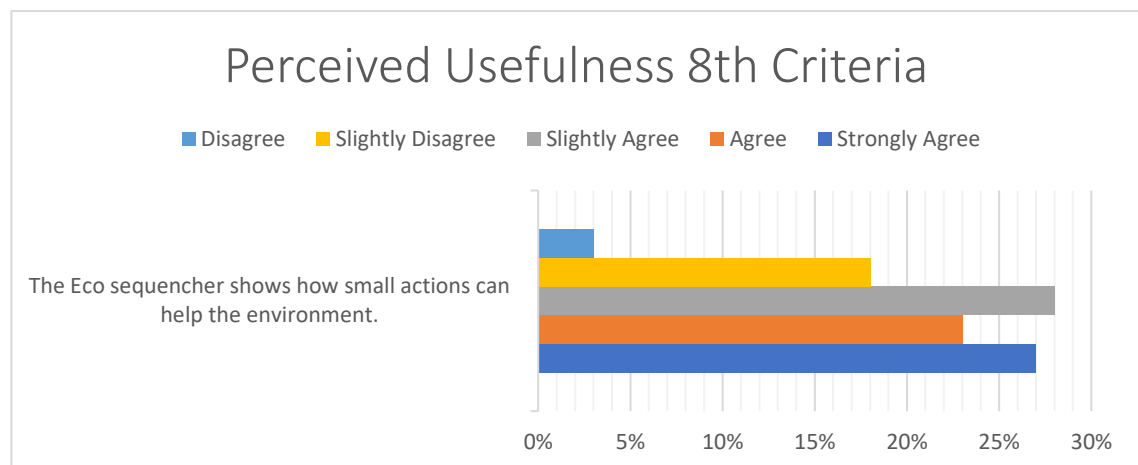


Figure 42 Perceived Usefulness 8th Criteria

The researchers integrated the use of bar infographics to effectively illustrate the data and to further enhance the visual comprehension of the key findings of the ratings of 90 respondents on the Eco-Sequencher System's perceived ease of use using a 6-point scale Technology Acceptance Model.

Table 10 shows the General mean of Perceived Ease which is 4.71

Table 11 General Mean of Perceived Ease of Use

Criteria	Rating N=92												
	6	%	5	%	4	%	3	%	2	%	1	%	GM
PERCEIVED EASE OF USE													
Students find the Eco-Sequencher's interface intuitive and user-friendly.	28	30%	33	36%	25	27%	4	4%	2	2%	0	0	4.88
Eco-Sequencher will be a big addition to schools' sustainability initiatives by providing hands-on education and promoting resource efficiency.	18	20%	28	30%	40	43%	4	4%	2	2%	0	0	4.60
Operating the Eco-Sequencher requires minimal time and effort.	26	28%	37	40%	21	23%	4	4%	4	4%	0	0	4.83
Setting up the Eco-Sequencher is easy.	17	18%	36	39%	32	35%	7	8%	0	0	0	0	4.68
The Eco-Sequencher instructions are helpful and easy to follow.	17	18%	35	38%	20	22%	19	21%	1	1%	0	0	4.52
The Instruction in LCD is easy to understand.	20	22%	34	37%	19	21%	17	18%	2	2%	0	0	4.55
The Eco-Sequencher is convenient for regular use.	27	29%	30	33%	29	32%	5	5%	1	1%	0	0	4.83
The Eco-Sequencher's design is simple and practical.	32	35%	24	26%	27	29%	6	7%	3	3%	0	0	4.82
GENERAL MEAN BASED ON PERCEIVED EASE OF USE													4.71

Eco-Sequencher

Where: 6 – Strongly Agree, 5 – Agree, 4 – Slightly Agree, 3 – Slightly Disagree, 2 – Disagree, 1 – Strongly Disagree

Figure 43 presents a bar infographic illustrating responses to the first criterion of perceived ease of use regarding the Eco-Sequencer. The results indicate that 30% of respondents strongly agree, 36% agree, and 27% slightly agree that students find the Eco-Sequencer's interface intuitive and user-friendly. Conversely, 4% of respondents slightly disagree, and 2% disagree with this statement. These findings suggest that most respondents affirm the Eco-Sequencer's interface as intuitive and user-friendly.

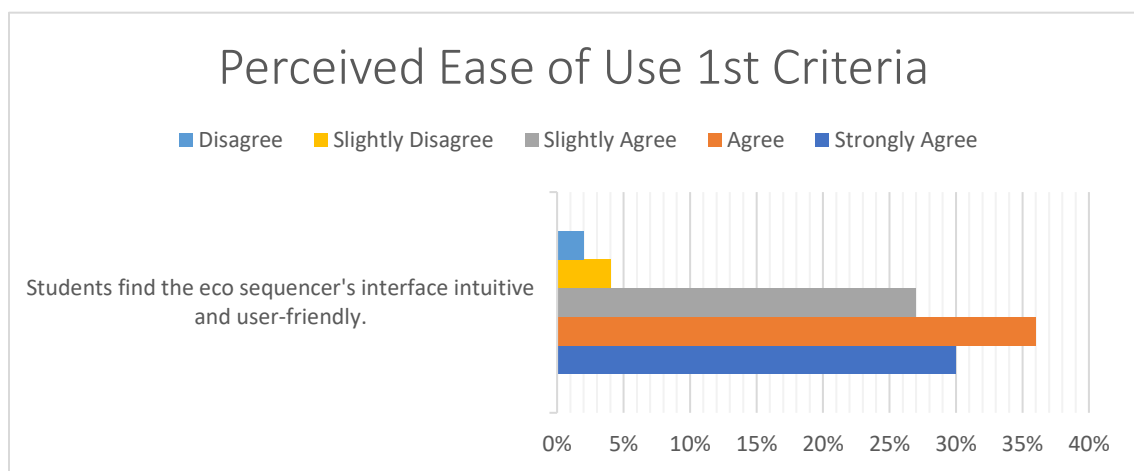


Figure 43 Perceived Ease of Use 1st Criteria

Figure 44 depicts the bar infographic for the second criterion of perceived ease of use. The data illustrates that 20% of respondents strongly agree, 30% agree, and 43% slightly agree that the Eco-Sequencer significantly enhances schools' sustainability initiatives by offering hands-on education and promoting resource efficiency. Conversely, 4% of respondents slightly disagree, and 2% disagree with this assertion. These results indicate that most respondents affirm the Eco-Sequencer's potential to be a substantial addition to schools' sustainability initiatives.

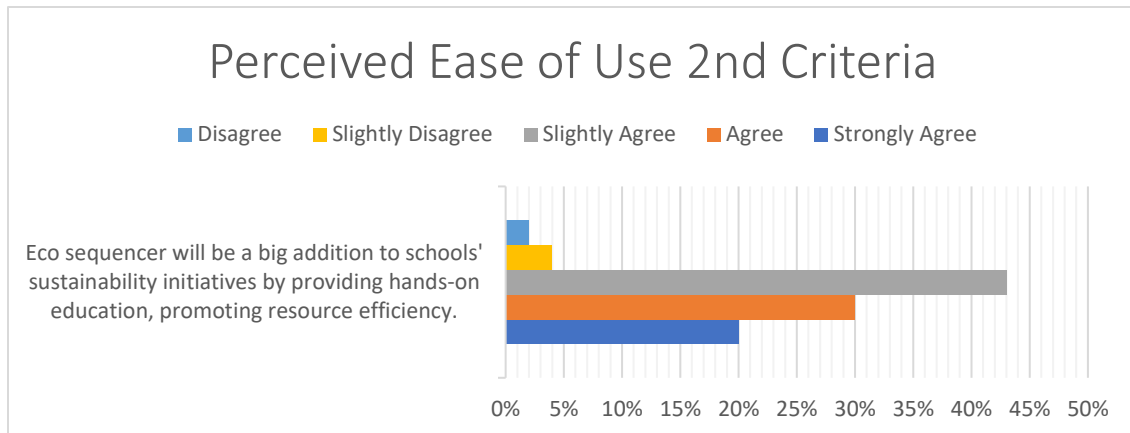


Figure 44 Perceived Ease of Use 2nd Criteria

Figure 45 illustrates the bar infographic for the third criterion of perceived ease of use. It indicates that 28% of respondents strongly agree, 40% agree, and 23% slightly agree that operating the Eco-Sequencer requires minimal time and effort, thus promoting resource efficiency. Conversely, 4% of respondents slightly disagree, and 4% disagree. These findings suggest that most respondents affirm the statement that operating the Eco-Sequencer demands minimal time and effort.

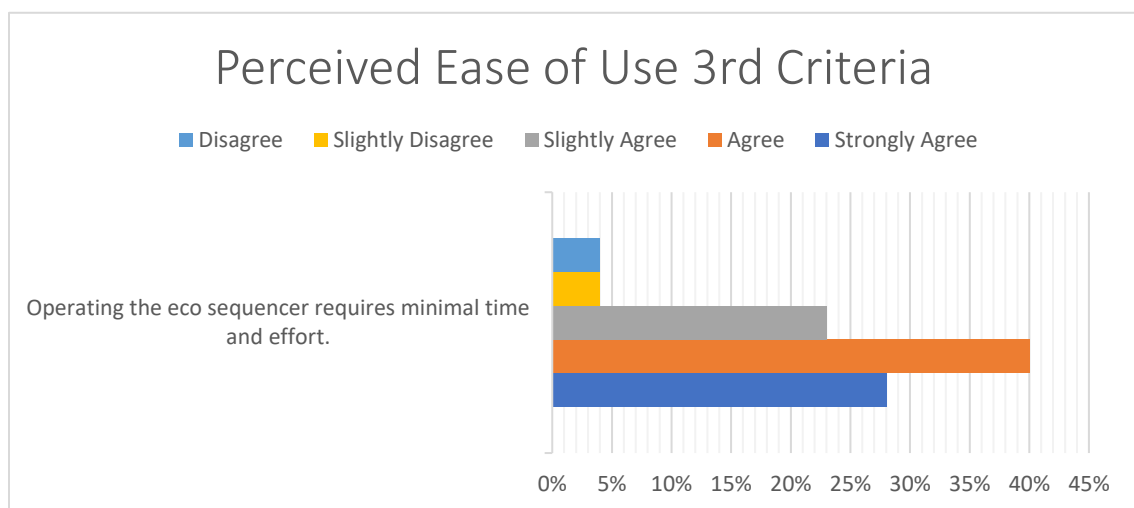


Figure 45 Perceived Ease of Use 3rd Criteria

Figure 46 presents the bar infographic for the fourth criterion of perceived ease of use, indicating that 18% of respondents strongly agree, 39% agree, and 35% slightly agree that setting up the Eco-Sequencher is easy. Conversely, 8% of respondents slightly disagree. These results suggest that most respondents affirm the ease of setting up the Eco- Eco-Sequencher.

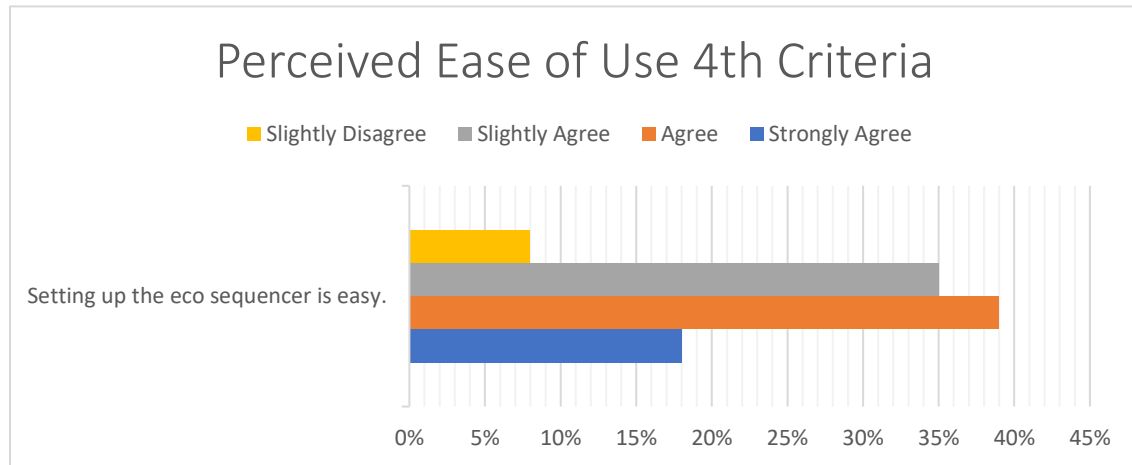


Figure 46 Perceived Ease of Use 4th Criteria

Figure 47 displays the bar infographic for the fifth criterion of perceived ease of use, revealing that 18% of respondents strongly agree, 38% agree, and 22% slightly agree that the Eco-Sequencher instructions are helpful and easy to follow. Conversely, 21% of respondents slightly disagree, and 1% disagree. These findings indicate that most respondents affirm the helpfulness and clarity of the Eco-Sequencher instructions.

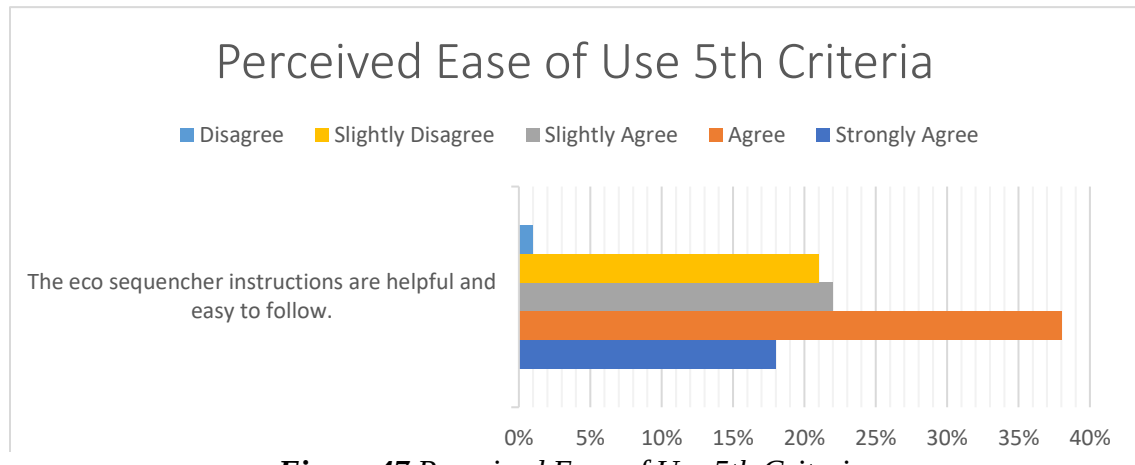


Figure 47 Perceived Ease of Use 5th Criteria

Figure 48 depicts the bar infographic for the sixth criterion of perceived ease of use, illustrating that 22% of respondents strongly agree, 37% agree, and 21% slightly agree that the instructions displayed on the LCD are easy to understand. Conversely, 18% of respondents slightly disagree, and 2% disagree. These results suggest that the majority of respondents affirm the clarity and comprehensibility of the instructions displayed on the LCD.

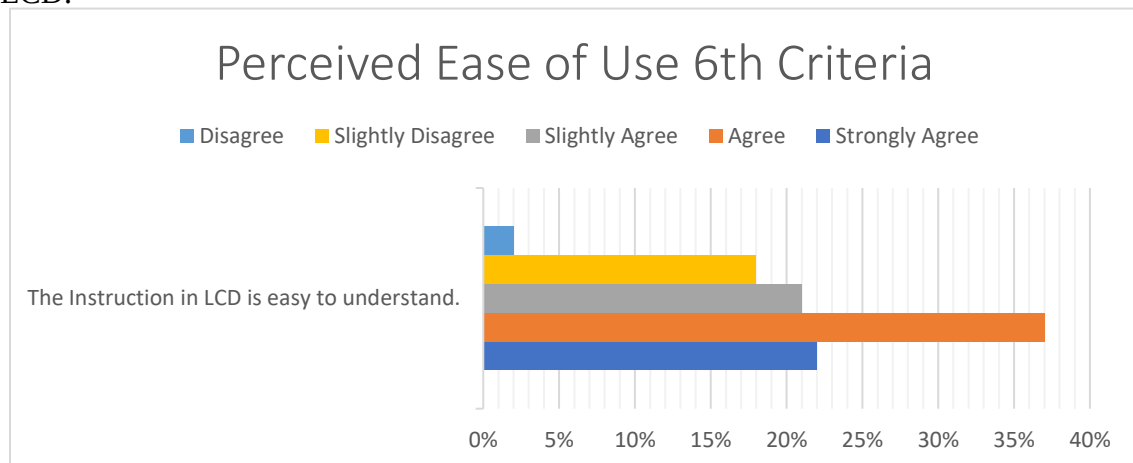


Figure 48 Perceived Ease of Use 6th Criteria

Figure 49 presents the bar infographic for the seventh criterion of perceived ease of use, indicating that 29% of respondents strongly agree, 33% agree, and 32% slightly agree that the Eco-Sequencher is convenient for regular use. Conversely, 5% of respondents

slightly disagree, and 1% disagree. These findings suggest that most respondents affirm the convenience of using the Eco-Sequencher regularly.

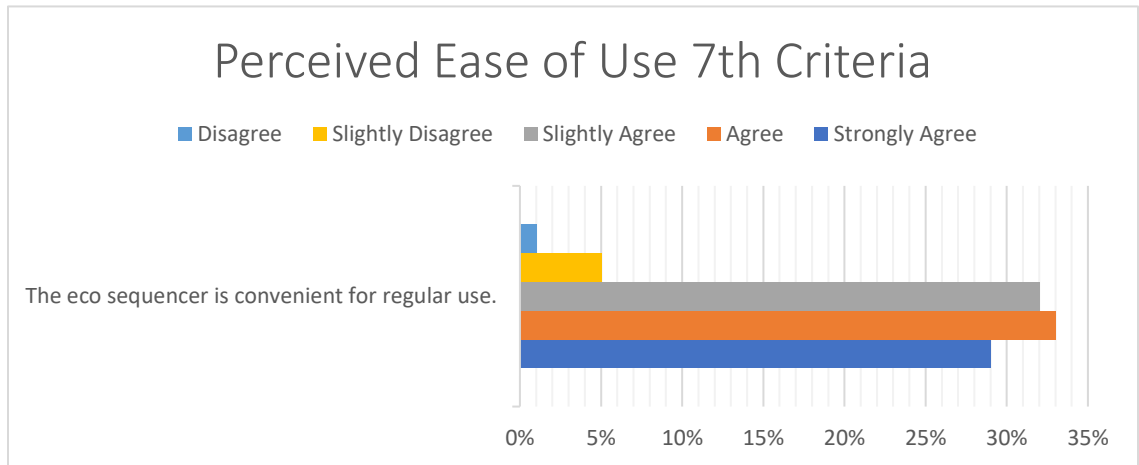


Figure 49 Perceived Ease of Use 7th Criteria

Figure 50 displays the bar infographic for the seventh criterion of perceived ease of use, revealing that 35% of respondents strongly agree, 26% agree, and 29% slightly agree that the Eco-Sequencher's design is simple and practical. Conversely, 7% of respondents slightly disagree, and 3% disagree. These results indicate that most respondents affirm the simplicity and practicality of the Eco-Sequencher's design.

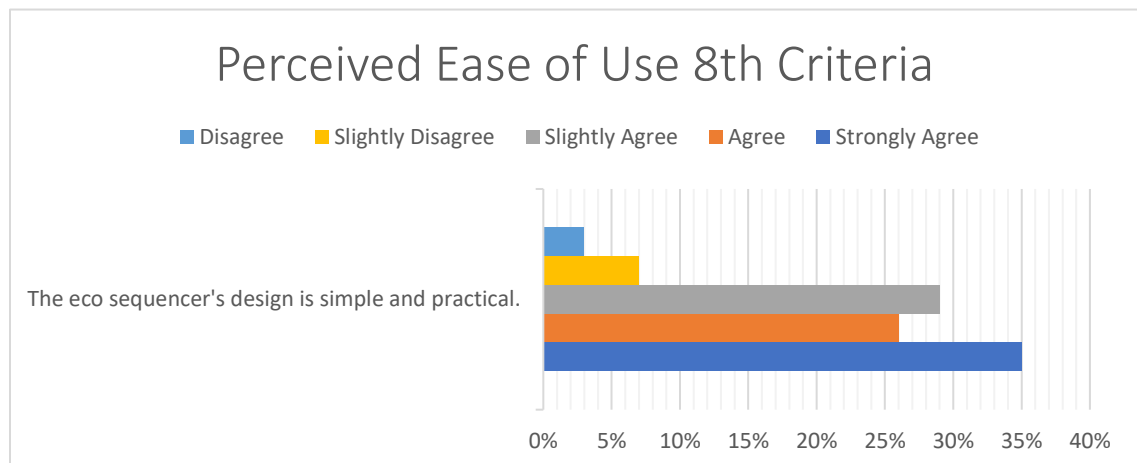


Figure 50 Perceived Ease of Use 8th Criteria

Chapter V

SUMMARY, CONCLUSION, AND RECOMMENDATION

The goal of this chapter is to offer a summary of the research. This document summarizes the procedures and methods used in the development of the Eco-Sequencher System, the outcome of the evaluation using the two primary factors of the Technology Acceptance Model, namely Perceived Usefulness and Perceived Ease of Use, conclusions based on the respondents' ratings using TAM, and recommendations for future research and the basis of innovation related to the study.

Summary

This study focuses on the Eco-Sequencher project, which is a reverse vending machine that requires consumers to insert a plastic bottle in exchange for clean and potable water. The system employs image processing to recognize the plastic bottle, a linear actuator to compress the plastic bottle to save space within the container, and a refilling mechanism to swap the plastic bottle for drinkable water. By collecting a plastic bottle, when the system accepts the user input, it is placed inside the machine container with the help of a sensor and servo for the mechanism, and provides a clean source of water with the help of a 3-stage filter and a UV light, ensuring its safety so it is fit to drink.

The development of the Eco-Sequencher involves the integration of both. Hardware and software components. The hardware involved in the hardware includes a camera that can identify plastic bottles with the help of image processing technology, an HX711 Load sensor for the identification of contents inside the water, a water flow sensor for the PWM (Pulse with Modulation) of the flow rate for water, 4x3 keypad membrane for getting user inputs, solenoid valve for closing the flow of water, i2c LCD for interacting

with the user, relay modules for automatic switching, LM2596 DC buck converter for lowering download voltage, Arduino Uno and Nano as the microcontroller, Push buttons for starting and cancelation, and Linear Actuator for compressing the plastic bottles, which are placed within the system to help the system identify plastic bottle in exchange to a water credit the can be exchanged to potable water. The software used is Arduino IDE which plays a vital role in the completion of the system it helps program the microcontroller unit to interact with the hardware components, Phyton IDE is also used for serial communication between the PC and MCU and it helps also on image processing technology and TinkerCAD act as the developing and optimizing the technology behind the system, from programming the control systems to designing the physical infrastructure that supports the development of the system.

The Eco-Sequencher offers several features and functions that contribute to its effectiveness in the proper collection of plastic bottles, ensuring the segregation of plastic bottles is applied, and offering clean, filter-potable water with the help of a water filtration system, even though the water comes from tap water. It employs a combination of a 3-stage filtration system and a UV light sterilizer to remove bad microorganisms, odors, and other particles that affect the impurity of the water for the safety of the user's health. When the intended user starts to put a plastic bottle in, the camera identifies it; if it is valid and has no content inside, it is accepted by the machine and exchanged for the required amount of water based on the weight of the bottle they input. These features and functions, allow the user to quench their thirst as well as help the environment by properly segregating plastic bottles.

Eco-Sequencher has received high satisfaction for its perceived usefulness and ease of use among engineering experts. This positive reception underscores Eco-Sequencher's role as a transformative solution in modern technology, effectively addressing technological demands and practical segregating needs. Integrating advanced sensor technology and filtration systems, Eco-Sequencher has been evaluated through the Technology Acceptance Model (TAM), highlighting its significant impact. The study combines qualitative methods and a systematic Project Development Life Cycle offering insights into the challenges and benefits of innovative hydroponic systems. The TAM survey yielded strong agreement, with a perceived usefulness rating of 4.67 and ease of use at 4.71, resulting in an overall score of 4.69. This score indicates strong endorsement from computer engineering experts regarding the project's benefits and effectiveness.

Conclusion

In conclusion, the device supported and resolved concerns of aiding students in practicing eliminating the use of plastic bottles as well as obtaining clean and free drinkable water through the usage of Eco-Sequencher by leveraging image processing and advanced sensor technologies such as the HX711 Load sensor and water flow sensor, Eco-Sequencher efficiently identifies and processes plastic bottles, ensuring only clean and empty containers are accepted for exchange with potable water. The integration of a 3-stage filtration system and UV light sterilizer further guarantees the safety and purity of the dispensed water, addressing both hydration needs and environmental sustainability by promoting proper plastic waste disposal and recycling. The project was evaluated using the two primary factors of the Technology Acceptance Model – perceived usefulness which gained a rating of 4.67 and perceived ease of use which gained a score of 4.71, resulting in

an overall score of 4.69 by the 90-student using Sloven's formula employing a 10% margin of error to ensure a representative sample from the total population of 942 students lower down to 90 and also 2 computer engineering experts were added as the respondents for the TAMP model. This concludes that the Eco-Sequencher gained a positive score feedback from 92 respondents using a 6-point scale rating. This means that the system offers a robust and transformative solution in recycling and water provision through its innovative reverse vending machine design, solving two problems the improper segregation of plastic bottles as well as gaining free access free potable water among the students of Eastwoods Professional College of Science and Technology Inc.

Recommendations

The recommendations of the researchers for future researchers working on projects comparable to the Eco-Sequencher System are included below.

1. **Advanced Sensor Integration:** Incorporate more advanced sensors to monitor the condition of the plastic bottles and the filtration process. This can help in improving the accuracy and efficiency of the system.
2. **Use Shredding Mechanism:** Upgrade to a shredding mechanism to handle a wider variety of plastic bottles and increase processing speed, thereby accommodating a larger volume of waste.

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Appendix A

INTERVIEW TRANSCRIPT

Our project is a machine with the capability to recycle or segregate plastic bottles in exchange for potable filtered clean water. For example, if you insert plastic bottles into our machine, it will identify if it is a plastic bottle and allow you to use the inserted plastic bottle in exchange for clean water using our machine. Additionally, you can store points using your student ID card to accumulate funds. If you need a source of clean water, you don't have to buy from retail stores anymore. This will quench your thirst and also help reduce the pollution caused by the improper disposal of plastic bottles, which is harmful to our environment.

1. Do you always consume potable water that is stored in a plastic bottle? If yes, do you buy it?

-Not really.

2. Do you think water should be free since it can be seen anywhere? If yes or no, why?

-It could be, on our campus, we need to have free water as a basic necessity.

3. Are you aware of the issue of plastic bottle pollution in our country? If yes, give your opinion about it.

-Yes, we really need to reduce the use of plastic bottles.

4. What do you think is a strategy we can use to raise awareness about the effects of plastic bottles and promote their exchange for clean water using the machine we propose?

-Absolutely.

5. In your opinion, will our proposed project help students get clean drinking water while also helping to segregate plastic bottles?

-It depends, but I think it's possible.

6. As an administrator of this school, would you want to implement this project if we get funding for it?

-If it turns out well, why not?

7. Do you also think it would help our maintenance to properly dispose of plastic bottles?

Yes, of course, because it will still be the students.

8. If our project is implemented in our school, will you use it instead of buying from outside? Why?

-That would be difficult, but we'll see if the school can provide the need.

9. As one of the managers of our school, do you think we can implement this project in our local barangay for it to become a project of our barangay officials?

-Because water is a need, I maintain that I think it cannot be done.

10. Give your opinion or suggestion about the problem that you want to be solved in our school.

My only suggestion is really about maintenance.


Ms. Rosalinda P. De Guzman
School Administrator

APPENDIX B

EVALUATION TOOL

Evaluation Tool for User Acceptability

Eco Sequencher

Respondent's profile

Name: _____

Position/ Profession: _____

Years of Service: _____ years

QUESTIONNAIRE

Please indicate a check mark (✓) under the column that best describes your responses for each item about the **Eco Sequencher**.

Please use the rating below:

- 6 – Strongly Agree
- 5 – Agree
- 4 – Slightly Agree
- 3 – Slightly Disagree
- 2 – Disagree
- 1 – Strongly Disagree

Please start answering here:
Technology Acceptance Model

Criteria	Ratings					
	6	5	4	3	2	1
PERCEIVED USEFULNESS						
1. The eco sequencher provides valuable benefits to the environment.						
2. Using the eco sequencher is a cost-effective way to obtain clean water.						
3. Having access to clean water through the Eco Sequencher promotes preventive healthcare.						
4. The eco sequencher is an eco-friendly device or machine.						

5. The eco sequencer is a reliable solution for segregating the waste.							
6. The presence of eco sequencers encourages students to engage in discussions about water quality and environmental issues.							
7. The eco sequencer helps students save money by giving them access to clean water just by using an empty bottle.							
8. The eco sequencer shows how small actions can help the environment.							
PERCEIVED EASE OF USE							
1. Students find the eco sequencer's interface intuitive and user-friendly.							
2. Eco sequencer will be big addition to schools' sustainability initiatives by providing hands-on education, promoting resource efficiency.							
3. Operating the eco sequencer requires minimal time and effort.							
4. Setting up the eco sequencer is easy.							
5. The eco sequencer instructions are helpful and easy to follow.							
6. The Instruction in LCD is easy to understand.							
7. The eco sequencer is convenient for regular use.							
8. The eco sequencer's design is simple and practical.							

Signature over printed name of Respondent

Date

APPENDIX C
Total Number of College students from 1st year to 4th year

EASTWOODS PROFESSIONAL COLLEGE OF SCIENCE AND TECHNOLOGY
ISAYO, CITY OF BALANGA BATAAN

COURSE	1st					2nd				3rd		4th		403	404	TOTAL
	101	102	103	104	105	201	202	203	204	301	302	401	402			
BSCPE	14					7				12		16	28			77
ACT	8					6										14
BSCS	5					9				7		9				30
BSIT	22	25	20			22	29	17		24	14	23	19	30		245
BSHM	26	19	27	13	19	21	16	19	10	26	28	19	29	14	10	296
HRS	20					16										36
HRT	25	24	16	20	23	8	6			23	8					153
Tourism	26	24														50
CET	21	20														41
																942

Prepared by:


Ms. Maria Deth A. Enriquez, LPT
 College Registrar

APPENDIX D

Certificate of Evaluation Tool Approval

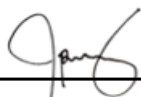
Certificate of Evaluation Tool Approval

This certifies that the evaluation tool developed by **JERVJ Group** has undergone rigorous assessment and has been found to meet all necessary criteria. After thorough evaluation and testing, it is confirmed that the said tool has passed and is deemed accurate.

This certification is issued on _____, to acknowledge the dedication and effort put forth by the creators of the evaluation tool. It signifies that the tool is approved for its intended use and is considered valid for conducting evaluations.

Approved by:

Signature: _____



Name: John Paul Marquez

Occupation: Odoo Functional Consultant

Certificate of Evaluation Tool Approval

This certifies that the evaluation tool developed by **JERVJ Group** has undergone rigorous assessment and has been found to meet all necessary criteria. After thorough evaluation and testing, it is confirmed that the said tool has passed and is deemed accurate.

This certification is issued on June 14, 2024, to acknowledge the dedication and effort put forth by the creators of the evaluation tool. It signifies that the tool is approved for its intended use and is considered valid for conducting evaluations.

Approved by:

Signature: _____



Name: Maren Joy Rutas

Occupation: Electronics Engineer

Certification from 2 Engineering Experts

APPENDIX E

Conversation From Engineering Experts

IPU 4:2 / PIV

Busy ka? Hahaha

Thu 5:23 PM

kagigising pa, nanuyta? hahaha

Tin ko mo pasagot questionnaire king thesis mi nung ok mo tas tin ko pa pirma haha

osige

Need mi kasi engineer eh haha aisip daka haha

➡ You forwarded a message

 ques.docx
17.77 KB

 Certificate-of-Evaluation-Tool-Approval1.docx
13.42 KB

← You replied to yourself

Attachment 

Ini papirma kayari hahah

🔙 You replied to yourself










The "Eco-Sequencher" is a reverse vending machine designed to encourage recycling by providing a sustainable reward system. This machine utilizes advanced image processing technology using machine learning to identify plastic bottles and a weight sensor to determine if the bottles are empty. If a bottle contains water, it will be rejected. Accepted bottles are compressed using a linear actuator to save space in the collection container.

Thursday 5:28pm

As an incentive, users are rewarded with clean, filtered, potable water. This water undergoes a thorough three-stage filtration process with filters of different micron levels to remove impurities. The final stage involves UV light sterilization to ensure the water is safe and fit for drinking. This system not only promotes recycling but also provides a valuable resource in the form of clean drinking water.

kapilan me kaylangan?

Ngeni or nung tin ka gagawan bukas namo

sige sige send ku nemu

Sge salamat

Thu 7:45 PM

ques.docx













to ensure the water is safe and fit for drinking. This system not only promotes recycling but also provides a valuable resource in the form of clean drinking water.

kapilan me kaylangan?

Ngeni or nung tin ka gagawan bukas namo

sige sige send ku nemu

Sge salamat

Thu 7:45 PM

ques.docx
40.12 KB

Certificate-of-Evaluation-Tool-Approval1.docx
35.34 KB

Thu 8:24 PM

Salamat ng marakal ne haha

Sent 1d ago

+ 📎 🗨️ GIF Aa 😊 👍

APPENDIX F

ACTUAL PHOTO OF ECO-SEQUENCHER DEPLOYMENT





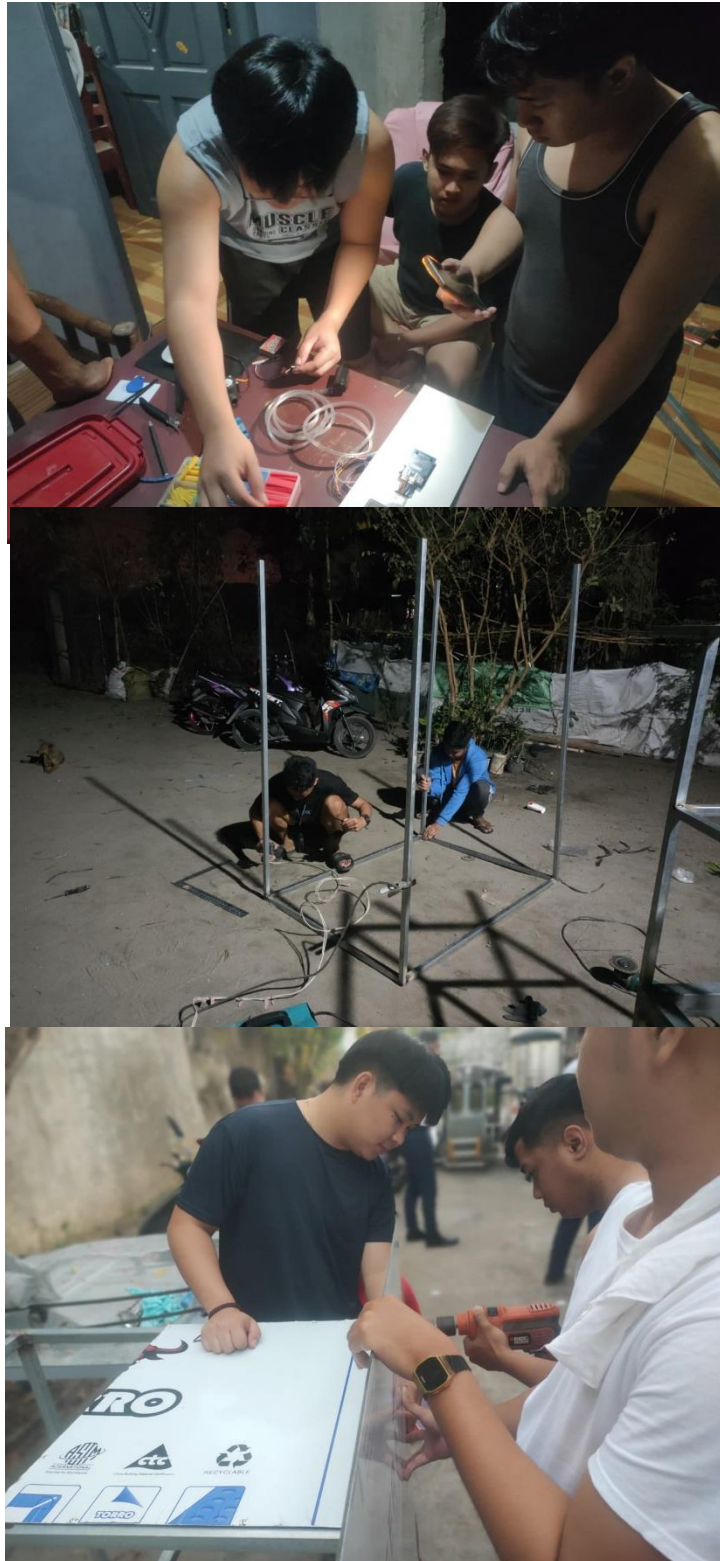
APPENDIX G

The Survey



APPENDIX H

Project Making







CURRICULUM VITAE



VINCENT R. FLORES

STUDENT

CAREER OVERVIEW

As an enthusiastic team player, I am eager to bring my positive attitude, strong work ethic, and unwavering dedication to excellence, along with my passion for delivering outstanding customer service, to create unforgettable experiences for all.

EDUCATION

Tertiary
2023-2024
BACHELOR OF SCIENCE IN COMPUTER ENGINEERING
Eastwoods Professional College of Science and Technology

2020-2023
DIPLOMA IN COMPUTER ENGINEERING
Eastwoods College of Science and Technology

Senior High School
2017-2019
COMPUTER PROGRAMMING
Eastwoods Professional College of Science and Technology

Junior High School
2013-2017
LUAKAN NATIONAL HIGH SCHOOL
Luacan, Dinalupihan Bataan

WORK EXPERIENCES AND TRAININGS

On-the-Job Training
ADDN+ The Architectural Firm
March-June 2022

UAQTEA (Tertiary Education Subsidy -TES)
Eastwoods College of Science and Technology
2020-2021

 09077764417

 TAGPISO420@GMAIL.COM

 #303 purok 2 Saguing,
Dinalupihan Bataan

SKILLS

- Basic Knowledge in Electronics and Mechatronics
- Basic Knowledge in Autodesk
- Organizational Skills
- Critical Thinking Skills

LANGUAGES

- English
- Filipino

REFERENCES

Marisol R. Flores
mariz0379@yahoo.com
09497628181
Patricia M. Jorge
Patricia_Jorge@gmail.com
0938-346-5832
Erwin T. Cortez
Erwin_Cortez03@gmail.com
0948-364-5548

I hereby certify that the information
stated above is true and correct.


VINCENT R. FLORES
Signature



📞 09152014953

✉️ jamespunzalan016@gmail.com

🏠 #255 Tucop, Dinalupihan
Bataan

SKILLS

- Basic Knowledge in Electronics and Mechatronics
- Basic Knowledge in Autodesk Autocadd
- Networking

LANGUAGES

- English
- Filipino

REFERENCES

Julie G. Punzalan
juliepunzalan@gmail.com
0954-079-7543
Jeremy F. Valdez
Jeremy_Valdez09@gmail.com
0921-543-5739

Henry D. Bautista
Henry_Bautistagmail.com
0927-160-1963

JAMES G. PUNZALAN

STUDENT

CAREER OVERVIEW

As a passionate collaborator, I thrive in dynamic environments where collaboration and innovation are key. My strong problem-solving abilities and ability to adapt quickly enable me to effectively address challenges and contribute to team success. I am committed to continuous learning and bringing creative solutions to every project

EDUCATION

Tertiary 2023-2024	BACHELOR OF SCIENCE IN COMPUTER ENGINEERING Eastwoods Professional College of Science and Technology
2020-2023	DIPLOMA IN COMPUTER ENGINEERING Eastwoods College of Science and Technology
Senior High School 2018-2020	STEM BASA AIR BASE NATIONAL HIGH SCHOOL
2014-2018	PAGALANGGANG NATIONAL HIGH SCHOOL Pagalanggang, Dinalupihan Bataan

WORK EXPERIENCES AND TRAININGS

On-the-Job Training
ADDN+ The Architectural Firm
March-June 2022

UAQTEA (Tertiary Education Subsidy -TES)
Eastwoods College of Science and Technology
2020-2021

I hereby certify that the information
stated above is true and correct.


JAMES G. PUNZALAN
Signature



 0946-384-5548

 jampol_Legaspi@gmail.com

 175 Balas, Almacen
Hermosa, Bataan

SKILLS

- Basic Knowledge in Electronics and Mechatronics
- Basic Knowledge in Autodesk and AutoCAD
- Networking
- Organizational Skills

LANGUAGES

- English
- Filipino

REFERENCES

Charriz Legaspi
charrizlegaspi@gmail.com
0985-971-3039

Teressa A. Punzalan
Terepunzalan12@gmail.com
0923-019-7432

Joselito A. Quinsey
Joselito_quinsey@gmail.com
0927-029-8472

JOHN PAUL LEGASPI

STUDENT

CAREER OVERVIEW

To leverage my computer engineering education and technical skills in software development and hardware design, and with a track of record of successfully completing projects, I am eager to contribute my skills and expertise to advance technological solutions in diverse industries.

EDUCATION

Tertiary
2023-2024
Bachelor of Science in Computer Engineering
Eastwoods Professional College of Science and Technology

2020-2023
Bachelor of Science in Computer Engineering
Eastwoods College of Science and Technology

Senior High School
2017-2019
General Academic Strand
Eastwoods College of Science and Technology

Junior High School
2013-2017
Balsik National High School
Balsik Hermosa, Bataan

WORK EXPERIENCES AND TRAININGS

On-the-Job Training
ADDN+ The Architectural Firm
March-June 2022

UAQTEA (Tertiary Education Subsidy -TES)
Eastwoods College of Science and Technology
2020-2021

I hereby certify that the information
stated above is true and correct.


JOHN PAUL LEGASPI
Signature



0948-662-1409

punggayfranco27@gmail.com

71 Lambingan St. Dampe,
Floridablanca, Pampanga

SKILLS

- Basic Knowledge in Electronics and Mechatronics
- Basic Knowledge in Autodesk
- Organizational Skills

LANGUAGES

- English
- Filipino

REFERENCES

Jocelyn Franco
asdvqesadtyge@gmail.com
0939-257-4564

Edessa A. Tuazon
Tuazon_Edessa@gmail.com
0927-129-6462

Reginald C. Villaflor
Reginald_Villaflor13@gmail.com
0948-369-9515

RONNIEL R. FRANCO

STUDENT

CAREER OVERVIEW

As an enthusiastic team player, I am eager to bring my positive attitude, strong work ethic, and unwavering dedication to excellence, along with my passion for delivering outstanding customer service, to create unforgettable experiences for all.

EDUCATION

Tertiary 2023-2024	Bachelor of Science in Computer Engineering Eastwoods Professional College of Science and Technology
2020-2023	Bachelor of Science in Computer Engineering Eastwoods College of Science and Technology
Senior High School 2017-2019	SCIENCE, TECHNOLOGY, ENGINEERING AND MATHEMATICS Basa Air Base National High School
Junior High School 2013-2017	Basa Air Base National High School Basa Air Base

WORK EXPERIENCES AND TRAININGS

On-the-Job Training
ADDN• The Architectural Firm
March-June 2022

UAQTEA (Tertiary Education Subsidy -TES)
Eastwoods College of Science and Technology
2020-2021

I hereby certify that the information
stated above is true and correct.

RONNIEL R. FRANCO
Signature



📞 09565932311

✉️ mendez.eugene10@gmail.com

🏠 89 Looban Layac, Dinalupihan Bataan

SKILLS

- Basic Knowledge in Electronics and Mechatronics
- Basic Knowledge in Autodesk
- Organizational Skills

LANGUAGES

- English
- Filipino

REFERENCES

Alicia R. Mendez

liciamendez196707@gmail.com
09569950887

Mary Angeline C. Dabu

Mary_angeline18@gmail.com

0948-369-9515

Jaycee F. Rivera

JCRivera09@gmail.com
0938-346-5832

EUGENE R. MENDEZ

STUDENT

CAREER OVERVIEW

As an enthusiastic team player, I am eager to bring my positive attitude, strong work ethic, and unwavering dedication to excellence, along with my passion for delivering outstanding customer service, to create unforgettable experiences for all.

EDUCATION

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2020-2023	Bachelor of Science in Computer Engineering Eastwoods College of Science and Technology
Senior High School 2017-2019	Tech-Voc Computer Programming Eastwoods College of Science and Technology
Junior High School 2013-2017	Luakan National High School-Annex San Ramon, Dinalupihan Bataan

WORK EXPERIENCES AND TRAININGS

On-the-Job Training

ADDN+ The Architectural Firm
March-June 2022

UAQTEA (Tertiary Education Subsidy -TES)

Eastwoods College of Science and Technology
2020-2021

I hereby certify that the information stated above is true and correct.


EUGENE R. MENDEZ
Signature